

**Ceylon Rail : AI Driven Intelligent Railway Operations
System for Sri Lanka**

INDIVIDUAL THESIS

Senawirathna D.M.N.T

(IT22151506)

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and Adaptive Seat Allocation Using AI

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DECLARATION AND SUPERVISOR CERTIFICATION

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Student name: Senawirathna D.M.N.T

Student ID: IT22151506

Signature: _____ Date: _____

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Role and name	Signature	Date
Supervisor: Ms. Chathurangika Kahandawaarachchi		
Co-supervisor: Udith Dharmakeerthi		

ABSTRACT

Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI

Railway transportation in Sri Lanka plays an important role in providing affordable and accessible public transportation; however, variations in passenger demand across travel periods, passenger classes, weekdays, seasons, and holidays create challenges in effective capacity management. Existing railway practices predominantly rely on predefined capacity allocation and operational rules, creating a gap between passenger-demand prediction and demand-aware seat allocation. This research presents an **Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation** component developed as an individual contribution to the CeylonRail Intelligence system, an AI-driven railway decision-support platform for Sri Lanka. The study aims to develop a data-driven approach that forecasts passenger demand and uses the predicted demand to support adaptive allocation of available seating capacity.

The research utilized historical railway ticketing and passenger-demand data covering **55,302 records from 2020 to 2024**. The methodology involved data preprocessing, exploratory demand analysis, temporal and seasonal feature engineering, machine-learning-based demand forecasting, and optimization-based seat allocation. **Random Forest and XGBoost** models were evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE), while an additional LSTM-based temporal forecasting experiment was conducted to investigate sequential demand patterns. Random Forest achieved the best performance among the primary forecasting models, with an **MAE of 24.10, RMSE of 37.67, and MAPE of 13.12%**. Class-wise evaluation further demonstrated differentiated prediction performance across first-, second-, and third-class passenger demand. An adaptive seat allocation mechanism was subsequently implemented using integer optimization to distribute capacity according to predicted demand while maintaining predefined allocation constraints.

The results demonstrate the feasibility of integrating passenger-demand forecasting with adaptive seat allocation to support more responsive and data-driven railway capacity management. The individual contribution provides a foundation for intelligent passenger-demand analysis and capacity allocation within the broader CeylonRail Intelligence system.

Keywords: Railway Demand Forecasting, Adaptive Seat Allocation, Machine Learning, Passenger Demand Prediction, Intelligent Railway Systems

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LIST OF ABBREVIATIONS

[Guidance - delete before submission: List abbreviations alphabetically. Define each abbreviation at first use in the main text even when it appears here.]

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma Separated Values
LSTM	Long Short Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
RMSE	Root Mean Squared Error
TCN	Temporal Convolutional Network
UI	User Interface
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall project context

Railway transportation is an important component of public transportation infrastructure, providing affordable and accessible mobility for large numbers of passengers. The effective operation of a railway network requires coordination among several interconnected factors, including passenger demand, train capacity, service reliability, scheduling, infrastructure constraints, and operational resources. As railway networks generate increasing volumes of operational and passenger-related data, Artificial Intelligence (AI), Machine Learning (ML), predictive analytics, and optimization techniques provide opportunities to transform conventional railway management into more data-driven and intelligent decision-making processes [1], [2].

The application of AI in railway transportation has expanded across areas such as passenger-flow prediction, train-delay prediction, railway traffic management, anomaly detection, timetable optimization, and operational decision support. Previous research has demonstrated that machine-learning and deep-learning techniques can identify complex patterns from historical transportation data and provide predictions that can support operational planning [3], [4]. However, the effectiveness of an intelligent railway system depends not only on the accuracy of individual predictive models but also on how their outputs are translated into practical operational decisions.

In Sri Lanka, railway transportation remains an important mode of public transportation and is particularly significant because of its affordability and extensive use by passengers travelling between major cities and regional areas. However, railway operations face several interconnected challenges. Passenger demand varies according to routes, travel periods, passenger classes, weekdays, seasonal conditions, and public holidays. These variations can create periods of overcrowding as well as periods of underutilized capacity. Consequently, effective passenger-demand analysis and capacity management are important for improving the efficiency and quality of railway services.

The modernization of railway operations through intelligent technologies therefore presents an opportunity to improve how railway authorities understand passenger demand and manage available resources. Rather than relying solely on historical averages or fixed operational rules, AI-based approaches can utilize historical data and contextual variables to identify demand patterns and provide predictive information for future decision-making.

The overall research project, **CeylonRail Intelligence: AI-Driven Intelligent Railway Operations System for Sri Lanka**, was developed to investigate this opportunity. The proposed system combines several intelligent railway capabilities within a unified decision-support framework. The overall project contains four major research components: passenger demand forecasting and adaptive seat allocation, train tracking and delay prediction, AI-based railway fraud detection and passenger

verification, and multi-objective train scheduling optimization.
AI_Intelligence_Railway_Operati...

The overall system is designed around a shared railway knowledge environment containing information such as railway routes, station networks, historical timetables, train capacities, passenger-demand information, ticketing policies, operational restrictions, maintenance information, and congestion-related information. This shared information environment allows outputs from one intelligent component to provide useful inputs to other components. For example, passenger-demand predictions can support capacity planning and scheduling, while train capacity information can be incorporated into seat-allocation decisions. AI_Intelligence_Railway_Operations

Therefore, the overall project aims to demonstrate how AI and data-driven technologies can be applied to address multiple operational challenges within the Sri Lankan railway context while providing a foundation for more intelligent and coordinated railway decision making.

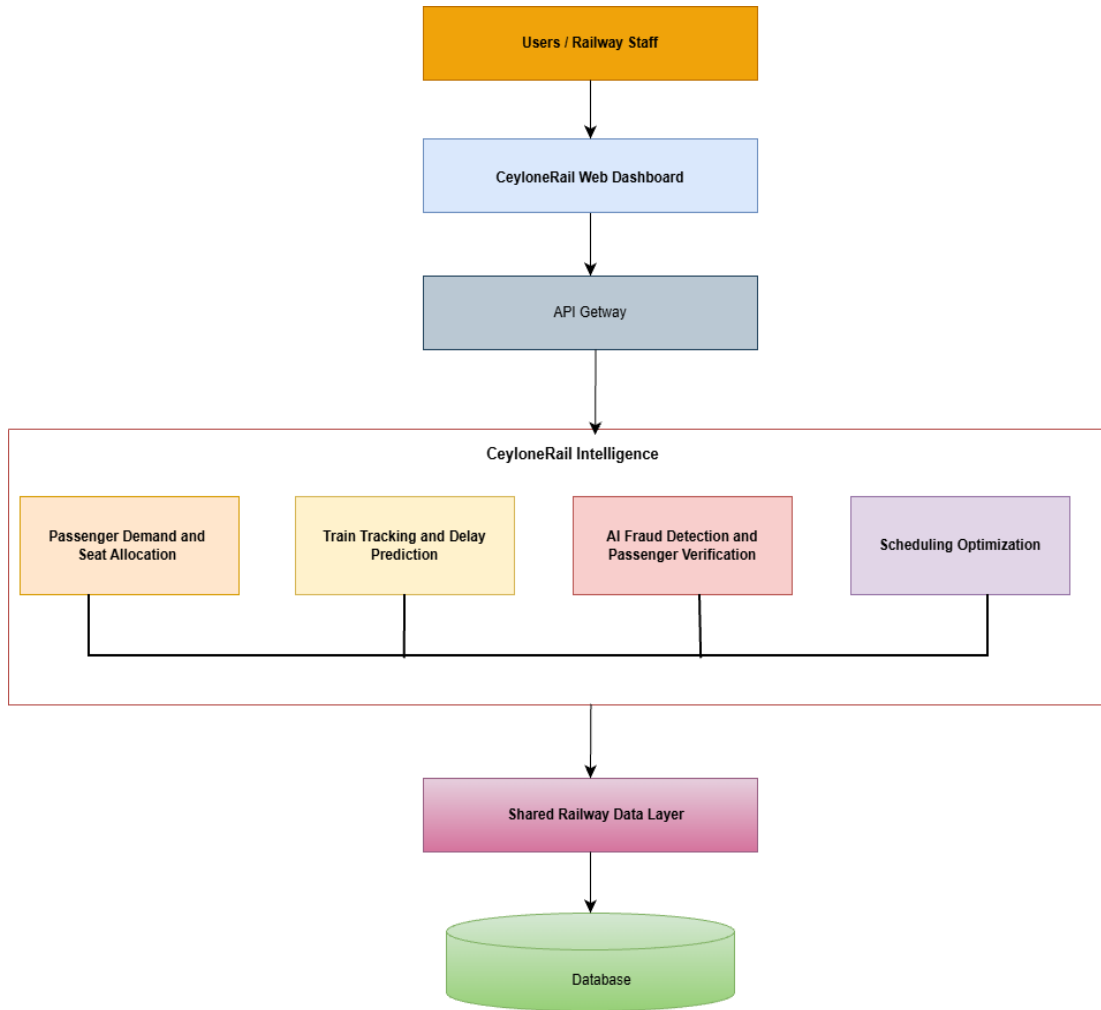


Figure 1.1 Overall Architecture of the CeylonRail Intelligence System

1.2 Individual component context

This individual thesis focuses on the **Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI** component of the CeylonRail Intelligence system.

Passenger demand is one of the fundamental factors influencing railway capacity planning. The number of passengers travelling on a particular service can vary substantially according to the route, passenger class, date, day of the week, month, public holidays, seasonal patterns, and other contextual factors. When capacity allocation does not adequately reflect these variations, some services or passenger classes may experience insufficient capacity while other capacity remains underutilized.

Passenger-demand forecasting provides an opportunity to address this problem by estimating future demand from historical observations and relevant contextual characteristics. However, forecasting passenger demand alone does not directly determine how available seats should be distributed. A practical railway capacity-

management approach must also translate demand predictions into allocation decisions while respecting the physical capacity of trains and predefined operational constraints.

The approved individual research component therefore focuses on two closely connected activities: **passenger-demand forecasting** and **adaptive seat allocation**. The Topic Assessment Form defines the component around collecting, integrating, and preprocessing railway ticketing data, calendar information, seasonal events, and route characteristics; developing and evaluating forecasting models; formulating adaptive seat allocation; and evaluating the resulting approach through controlled experiments.

The implemented research uses historical railway data covering the **2020–2024 period**. The dataset contains **55,302 records** and includes passenger-demand information associated with different railway travel classes and contextual variables. The implemented preprocessing and feature-engineering process incorporates temporal, calendar, route, train, capacity, holiday and historical demand-related characteristics.

The individual component consequently follows a two-stage process. In the first stage, historical passenger information is processed and analyzed to identify demand patterns and develop demand forecasts. In the second stage, predicted demand is used to support adaptive allocation of available seating capacity among passenger classes.

This individual component therefore provides the connection between passenger-demand intelligence and railway capacity management within the broader CeylonRail Intelligence platform.

1.3 Problem statement

Passenger demand within railway networks is not constant. Demand can change according to travel dates, weekdays, seasons, public holidays, routes, passenger classes, and other operational or contextual factors. Consequently, allocating railway seating capacity using fixed or predetermined approaches may not adequately respond to changing passenger requirements.

In the Sri Lankan railway context, the availability of historical passenger and ticketing information creates an opportunity to use data-driven methods to understand these demand patterns. However, converting historical records into reliable predictions and subsequently using those predictions to make capacity-allocation decisions remains a challenging problem.

Existing approaches can address individual aspects of this problem. A forecasting model can estimate future passenger demand, while a separate allocation mechanism can distribute available capacity. However, treating these activities independently can limit the operational value of the prediction. A demand forecast becomes more useful when it can directly support a subsequent capacity-management decision.

The problem addressed by this research can therefore be expressed as follows:

How can historical railway passenger-demand data be used to develop an AI based demand forecasting approach and subsequently integrate its predictions into an adaptive seat allocation mechanism that responds to changing demand while respecting railway capacity constraints?

This problem contains two interconnected technical challenges. The first is the development of a forecasting approach capable of learning meaningful demand patterns from historical railway data. The second is the development of an allocation mechanism capable of translating predicted demand into practical seat-allocation decisions.

The research therefore does not consider passenger-demand forecasting as an isolated prediction problem. Instead, it investigates the integration of **prediction and capacity allocation** as a combined decision-support process.

1.4 Research gap

The increasing application of AI and machine learning to railway transportation has resulted in substantial research into passenger-flow prediction, delay prediction, railway traffic management, and other intelligent railway applications. Existing research demonstrates the ability of statistical and machine-learning approaches to model transportation demand and identify temporal patterns [3]–[5]. Deep learning approaches such as Long Short-Term Memory (LSTM) networks have also been applied to sequential transportation prediction because of their ability to learn temporal dependencies in historical observations [6].

However, several gaps remain relevant to the Sri Lankan railway context.

First, much of the existing railway-demand forecasting research focuses primarily on the **prediction problem**. Although forecasting future passenger demand can provide useful information prediction by itself does not necessarily result in an operational improvement. The predicted demand must be connected to a practical railway management decision such as capacity allocation.

Second, existing railway systems and operational practices commonly rely on predetermined timetables, fixed capacity arrangements, rule based processes or manually supported decision making. The approved research proposal identifies the limited integration between forecasting and operational optimization as an important limitation within existing approaches.

Third, railway demand is influenced by multiple temporal and contextual factors. A forecasting approach designed for a particular environment may not necessarily transfer directly to the Sri Lankan railway context because passenger behavior, route characteristics, travel patterns and operational practices can differ between countries and railway networks.

Fourth, there is a need to consider capacity allocation after demand prediction. Even when a forecasting model provides an estimate of future demand, the available seats remain constrained by the physical capacity of the railway service. Therefore, a useful intelligent system must consider both predicted demand and allocation constraints when making capacity decisions.

The approved individual research component addresses these gaps by proposing a **Sri Lanka-specific demand forecasting and adaptive seat allocation framework**. The intended contribution is not simply to predict passenger demand, but to couple demand predictions with an adaptive allocation mechanism that considers demand, available capacity, passenger class requirements, and allocation constraints. The TAF identifies this integration as the main novelty of the individual component.

Accordingly, the research gap addressed by this thesis can be summarized as the lack of a sufficiently integrated, Sri Lankan railway specific approach that connects data driven passenger-demand forecasting with adaptive railway seat allocation.

1.5 Aim and objectives

1.5.1 Research Aim

The aim of this individual research is:

To design, implement, and evaluate an AI-based passenger-demand forecasting and adaptive seat allocation approach that supports demand-aware railway capacity management in the Sri Lankan railway context.

The research seeks to achieve this aim by analyzing historical railway passenger data, identifying relevant temporal and contextual demand patterns, evaluating machine-learning forecasting approaches, and integrating predicted demand into an adaptive seat-allocation mechanism.

1.5.2 Research Objectives

The following objectives were defined for the individual research component.

Objective 1 – Data Collection, Integration and Preprocessing

To collect, integrate, and preprocess relevant historical railway ticketing and passenger-demand data together with calendar, seasonal, route, train, and capacity-related information to create a suitable dataset for demand analysis and forecasting.

Objective 2 – Passenger-Demand Forecasting

To develop and evaluate machine-learning and temporal forecasting approaches capable of modelling passenger-demand patterns using historical railway data and relevant contextual features.

Objective 3 – Adaptive Seat Allocation

To develop an adaptive seat-allocation mechanism that uses predicted passenger demand to determine class-wise seat allocation while satisfying available capacity and predefined allocation constraints.

Objective 4 – Evaluation and Validation

To evaluate the forecasting and adaptive seat allocation approaches using appropriate prediction and capacity management metrics and controlled experimental analysis.

These objectives are derived from the approved individual component objectives in the Topic Assessment Form.

1.6 Research questions or hypotheses

Based on the research problem, identified gap, aim, and objectives, this research addresses the following research questions.

RQ1

How can historical railway passenger and contextual data be processed to identify temporal, seasonal and class level patterns that can support passenger demand forecasting?

RQ2

How effectively can machine learning and temporal forecasting approaches predict passenger demand using historical Sri Lankan railway data?

RQ3

How can predicted passenger demand be incorporated into an adaptive seat allocation mechanism while satisfying railway capacity and allocation constraints?

RQ4

To what extent can demand based seat allocation improve the utilization of available seating capacity across different passenger classes?

RQ5

What are the practical limitations and considerations associated with applying AI based demand forecasting and adaptive seat allocation within the Sri Lankan railway environment?

These questions establish a direct relationship between the research problem, experimental methodology and evaluation presented in the subsequent chapters.

1.7 Scope limitations and assumptions

1.7.1 Scope

The scope of this individual research is limited to passenger demand forecasting and adaptive seat allocation within the wider CeylonRail Intelligence system.

The research includes:

- Historical railway passenger and ticketing data analysis.
- Data cleaning and preprocessing.
- Feature engineering.
- Temporal and seasonal demand analysis.
- Passenger demand forecasting.
- Comparison of implemented forecasting models.
- Class-wise demand analysis.
- Demand based adaptive seat allocation.
- Railway capacity constraints.
- Class wise seat utilization analysis.
- Evaluation using forecasting error metrics.
- Evaluation of the relationship between predicted demand and allocated capacity.
- Integration of the component within the broader CeylonRail Intelligence architecture.

The implemented dataset covers railway records from 2020 to 2024 with the forecasting experiments using historical observations for model development and 2024 observations for testing.

The individual research does not attempt to independently develop the other major CeylonRail Intelligence components including train delay prediction, railway fraud detection and passenger verification or multi objective train scheduling optimization. These are separate research contributions within the overall group project.

1.7.2 Limitations

Several limitations should be considered when interpreting the findings of this research.

First, the performance of the forecasting models depends on the quality, completeness, and representativeness of the available historical railway data. Missing information, inconsistencies or limited coverage of particular operational situations may affect model performance.

Second historical passenger behaviour does not necessarily represent all future travel behavior. Unexpected events, changes in railway operations, major disruptions, changes in passenger preferences, and unusual travel periods may result in demand patterns that differ from historical observations.

Third, the forecasting experiments are based on the available dataset and selected features. Additional real-time information, such as live booking activity, real-time train occupancy, weather conditions, large-scale events, or real-time railway disruptions, could potentially provide additional predictive information but was outside the implemented scope.

Fourth, the adaptive seat allocation mechanism is evaluated using the capacity and demand information available within the developed system. The experimental results therefore demonstrate the feasibility of the approach but should not automatically be interpreted as a direct replacement for operational railway allocation policies without further real-world validation.

Finally, the research evaluates the developed approach under controlled experimental conditions. Large-scale deployment would require further testing using continuously updated operational railway data and collaboration with railway authorities.

1.7.3 Assumptions

The research is based on several assumptions.

It is assumed that historical railway passenger demand data contains meaningful patterns that can be used to estimate future demand.

It is also assumed that temporal, seasonal, calendar, route, train, class and historical demand characteristics contain information that can contribute to demand prediction.

The adaptive seat allocation component assumes that the available train capacity is known or can be represented within the system and that predicted passenger demand can be used as an input for allocation decisions.

Finally, it is assumed that the constraints defined within the allocation model reasonably represent the operational requirements of the experimental environment.

1.8 Individual contribution

The primary individual contribution of this research is the development and evaluation of the Advanced Spatio Temporal Demand Forecasting and Adaptive Seat Allocation using AI component within the CeylonRail Intelligence system.

The contribution begins with the preparation and analysis of historical railway passenger demand data. The implemented data-processing pipeline incorporates relevant temporal, calendar, route, train, capacity, holiday and historical demand characteristics to support demand analysis and forecasting.

The research further contributes a machine learning-based demand forecasting process and evaluates forecasting performance using quantitative error measures including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). The implemented experiments include Random Forest and XGBoost forecasting approaches, together with an LSTM based temporal forecasting experiment for investigating sequential demand patterns.

The second major contribution is the development of an adaptive seat allocation mechanism that uses predicted passenger demand as an input to capacity allocation. The implementation formulates seat allocation as an optimization problem while considering train capacity and predefined class level allocation constraints.

The contribution is therefore not limited to developing a prediction model. It establishes a connection between passenger demand prediction and operational capacity allocation, allowing forecast information to influence the distribution of available seats.

Individual contribution supports the broader CeylonRail Intelligence system by providing passenger-demand intelligence that can potentially support other railway decision making processes. Within the overall project, other research members address train-delay prediction, railway fraud detection and passenger verification, and multi objective train scheduling optimization. The individual technical responsibility of this thesis remains focused on passenger-demand forecasting and adaptive seat allocation.

The approved Topic Assessment Form identifies the Sri Lankan railway specific forecasting framework coupled with an adaptive, fairness aware seat allocation mechanism as the principal novelty of this individual component.

1.9 Thesis structure

This thesis is organized into nine chapters.

Chapter 1 – Introduction introduces the overall CeylonRail Intelligence project and the individual research component. It presents the research context, problem statement,

research gap, aim, objectives, research questions, scope, limitations, assumptions and individual contribution.

Chapter 2 – Literature Review and Related Work reviews existing research and theoretical foundations related to railway passenger-demand forecasting, time series forecasting, machine learning, deep learning, adaptive capacity allocation, and intelligent railway systems. The chapter evaluates existing approaches and synthesizes the research gap addressed by this study.

Chapter 3 – Research Methodology presents the research design, dataset, data-preprocessing procedures, feature engineering process, forecasting methodology, adaptive seat allocation methodology, experimental procedure, evaluation metrics, ethical considerations, and reproducibility practices.

Chapter 4 – Product and Solution Design describes the design of the individual component within the CeylonRail Intelligence system. It presents the system architecture, component interactions, forecasting workflow, adaptive seat allocation design, data flow, constraints and major design decisions.

Chapter 5 – Implementation, Integration and Testing presents the development environment, technologies, data-processing implementation, forecasting implementation, adaptive allocation implementation, integration with the broader system, software testing, and deployment considerations.

Chapter 6 – Results and Validation presents the experimental findings of the research. This includes demand analysis results, forecasting performance, class wise forecasting results, adaptive seat allocation results, seat-utilization analysis, technical validation, and integration results.

Chapter 7 – Discussion interprets the findings and relates them to the research objectives and existing literature. It discusses the effectiveness of the proposed approach, individual research contribution, technical contribution, limitations, threats to validity and practical implications.

Chapter 8 – Commercialization and Sustainability discusses the potential value of the developed solution, target users, market and competitive considerations,

sustainability, scalability, compliance, and its potential contribution to relevant Sustainable Development Goals.

Chapter 9 – Conclusion and Future Directions summarizes the research findings, evaluates the achievement of the research objectives, presents the individual research and product contributions and identifies possible directions for future development.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was conducted to establish the theoretical and technical foundation for the development of the **Advanced Spatio Temporal Demand Forecasting and Adaptive Seat Allocation using AI** component of the CeylonRail Intelligence system. The review focused on existing research and technological approaches related to railway passenger demand forecasting, time-series prediction, machine learning, deep learning, spatio temporal demand modelling, railway capacity management, and adaptive seat allocation.

The review primarily considered peer reviewed journal articles, conference papers, and relevant technical literature addressing transportation demand forecasting and intelligent railway operations. Particular attention was given to studies applying statistical forecasting, machine-learning algorithms, and deep-learning architectures to passenger demand prediction. Research concerning capacity allocation and optimization was also considered because the individual component extends demand forecasting into an operational seat-allocation decision.

The review was structured around five main themes. The first theme examined **railway passenger-demand characteristics**, including temporal, seasonal, weekday, holiday, route, and passenger class variations. The second examined **forecasting techniques**, including conventional statistical approaches and machine learning and deep-learning methods. The third focused on **spatio temporal demand forecasting**, where demand is influenced by both temporal patterns and spatial or route-related characteristics. The fourth examined **railway capacity and seat allocation approaches**, including rule based and optimization-based techniques. The fifth considered the integration of forecasting outputs with operational decision making.

The scope of the review was deliberately aligned with the individual research objectives. Studies concerned primarily with train-delay prediction, railway fraud detection, passenger verification, or multi objective timetable optimization were considered outside the main scope unless they provided relevant background to the overall CeylonRail Intelligence system. The review therefore concentrates on

literature that can directly inform the design, implementation, and evaluation of passenger-demand forecasting and adaptive seat allocation.

The findings from the reviewed literature are subsequently used to identify limitations in existing approaches and establish the research gap addressed by this thesis. The review also provides the theoretical basis for selecting appropriate forecasting techniques, allocation strategies, and evaluation measures for the proposed individual component.

2.2 Domain and theoretical foundations

2.2.1 Railway Passenger Demand

Passenger demand represents the number of passengers who require railway transportation during a particular period. In railway systems demand can vary according to travel date, time of day, day of the week, passenger class, route, season and public holidays. Understanding these variations is important because railway capacity is limited and cannot always be increased immediately when demand increases.

For this research passenger demand is treated as a measurable time dependent variable that can be learned from historical railway records. The dataset used in the implementation contains passenger ticketing information from 2020 to 2024. The analysis considers demand across different travel classes together with temporal and operational characteristics. This provides a foundation for identifying recurring demand patterns and developing forecasting models.

Passenger demand forecasting is particularly important for capacity management because predicted demand can provide information about where and when additional capacity may be required. Research in transportation forecasting has shown that demand can contain both temporal and spatial relationships. These relationships become more complex when external factors are also considered.

2.2.2 Time Series Forecasting

Time series forecasting is the process of using historical observations to estimate future values of a variable. Unlike ordinary prediction problems time series forecasting

considers the ordering of observations because previous observations may contain information about future observations.

Railway passenger demand can be treated as a time series because passenger volumes are recorded across successive dates. Important characteristics can include trends, seasonal patterns and short term fluctuations. Traditional statistical approaches such as ARIMA model relationships between current observations and previous observations. ARIMA based approaches have been widely used in demand forecasting and railway ticket demand research.

The effectiveness of time series forecasting depends on the characteristics of the available data. When passenger demand contains nonlinear relationships or depends on multiple contextual variables machine learning methods can provide an alternative to purely statistical approaches.

2.2.3 Machine Learning for Demand Forecasting

Machine learning enables predictive models to learn relationships between input variables and a target variable from historical data. In passenger demand forecasting the input variables can include historical demand, calendar information, passenger class, route characteristics, train characteristics and other contextual variables.

Tree based models such as Random Forest can learn nonlinear relationships between multiple input features and passenger demand. Random Forest combines multiple decision trees and produces predictions by aggregating the outputs of individual trees. This makes it suitable for datasets containing a mixture of numerical and categorical characteristics.

Gradient boosting approaches such as XGBoost provide another machine learning approach for structured prediction problems. These models build an ensemble of decision trees sequentially where later trees attempt to improve errors produced by previous trees. Such approaches can be useful when demand depends on complex interactions between different features.

The implementation of this research evaluates Random Forest and XGBoost as the primary machine learning forecasting approaches. Their performance is evaluated using MAE, RMSE and MAPE.

2.2.4 Deep Learning and LSTM

Deep learning is a branch of machine learning that uses neural network architecture with multiple processing layers. Deep learning approaches are particularly relevant to sequential data because they can learn complex relationships from historical observations.

Long Short Term Memory networks are a type of recurrent neural network designed to learn dependencies across sequences. LSTM networks use memory cells and gating mechanisms to control which information should be retained or discarded. This characteristic makes LSTM suitable for problems where previous observations can influence future values.

LSTM has been applied to passenger demand forecasting in transportation systems. Research has demonstrated that LSTM based models can capture temporal relationships in passenger demand. Other studies have combined LSTM with spatial modelling techniques to represent both spatial and temporal dependencies in passenger volume prediction.

In this research an LSTM based forecasting experiment was also implemented to investigate sequential passenger demand behavior. The LSTM experiment uses historical demand observations to learn temporal patterns and produces predicted demand values for evaluation.

2.2.5 Spatio Temporal Demand Forecasting

Spatio temporal forecasting considers both **where** and **when** an event occurs. In transportation systems passenger demand can vary between locations while also changing overtime. Therefore a forecasting model that considers only historical demand values may not capture all relevant demand relationships.

Railway passenger demand can contain spatial characteristics through routes, stations, origin destination relationships and travel classes. At the same time demand can contain temporal characteristics through daily patterns, weekly patterns, monthly patterns, seasonal behavior and holiday effects.

Research in transportation demand forecasting has demonstrated the importance of combining spatial and temporal information. Deep learning approaches have been developed to capture spatial relationships and temporal dependencies simultaneously.

The individual research component applies this principle by incorporating route related characteristics together with temporal and historical demand features. This provides a foundation for analyzing demand patterns within the Sri Lankan railway environment.

2.2.6 Adaptive Seat Allocation

Seat allocation refers to determining how available seating capacity should be distributed among passengers or passenger classes. In railway systems the number of available seats is constrained by the physical capacity of a train. Therefore seat allocation becomes an optimization problem when demand differs between passenger classes or travel segments.

Traditional allocation approaches may use fixed rules or predetermined capacity ratios. Such approaches may not respond effectively when demand changes over time. Adaptive seat allocation instead uses current or predicted demand information to determine how available capacity should be distributed.

Existing railway research has formulated seat allocation as an optimization problem. For example, research on passenger railway systems has developed mathematical optimization models for determining seat allocation while considering demand and capacity constraints.

In the proposed research demand predictions are used as an input to the seat allocation process. The allocation mechanism considers predicted demand together with available capacity and predefined class allocation constraints.

2.2.7 Optimization Based Capacity Allocation

Optimization is the process of selecting the best feasible solution according to a defined objective while satisfying a set of constraints. In railway seat allocation the objective can involve maximizing satisfied demand, improving capacity utilization or balancing different passenger requirements.

Mixed Integer Linear Programming has been used for railway seat allocation and capacity management because allocation decisions can be represented using integer decision variables and mathematical constraints. Research has also investigated stochastic optimization to account for uncertain passenger demand.

The implementation of this research formulates adaptive seat allocation as an optimization problem. The model considers predicted demand and train capacity while applying minimum and maximum allocation constraints for different passenger classes. This allows the allocation process to respond to predicted demand rather than relying entirely on a fixed allocation arrangement.

2.2.8 Fairness in Seat Allocation

Fairness is an important consideration when available railway capacity is distributed among different passenger classes or travel groups. A system that maximizes utilization without considering allocation constraints could allocate a disproportionate amount of capacity to one group.

Recent railway research has explicitly considered equity in seat allocation. Research on railway seat allocation with equity considerations has formulated optimization models that attempt to distribute capacity more fairly among different origin destination passenger groups while also considering operational objectives.

The approved individual research component also identifies fairness as an important aspect of the proposed adaptive allocation approach. In the implemented system fairness related allocation is represented through predefined class allocation constraints. These constraints provide a controlled mechanism for preventing the optimization process from allocating capacity without considering class level requirements.

2.2.9 Forecasting Evaluation Metrics

Forecasting performance needs to be evaluated using quantitative measures. This research uses Mean Absolute Error, Root Mean Square Error and Mean Absolute Percentage Error.

Mean Absolute Error (MAE) measures the average absolute difference between actual demand and predicted demand. Lower MAE indicates that predictions are closer to the actual observations.

Root Mean Square Error (RMSE) gives greater influence to larger prediction errors because the errors are squared before the average is calculated and the square root is then taken. It is therefore useful when larger forecasting errors are particularly important.

Mean Absolute Percentage Error (MAPE) expresses forecasting error as a percentage of the actual value. It provides an interpretable measure of relative prediction error although it can become problematic when actual values are zero or very close to zero.

These metrics are used in this research to compare forecasting performance and determine the effectiveness of the implemented demand forecasting approaches.

2.2.10 Relationship Between Forecasting and Seat Allocation

The central theoretical foundation of this research is the connection between demand forecasting and capacity allocation. Forecasting provides an estimate of future passenger requirements while optimization provides a mechanism for translating that estimate into an operational decision.

2.3 Related research

Research on railway passenger demand forecasting has developed from conventional statistical forecasting towards machine learning and deep learning approaches capable of representing complex temporal and spatial patterns. At the same time research on railway seat allocation has mainly focused on capacity management revenue management passenger choice and optimization. However the integration of demand forecasting with adaptive seat allocation remains comparatively limited particularly in the context of Sri Lankan railway operations.

2.3.1 Traditional Passenger Demand Forecasting

Traditional statistical forecasting techniques such as ARIMA have been widely used for transportation demand prediction because of their ability to model temporal dependencies in historical observations. These approaches are useful when demand

follows relatively stable temporal patterns and when sufficient historical observations are available. However railway passenger demand can be affected by several factors simultaneously including day of the week holidays seasonal conditions route characteristics train type and passenger class. This creates challenges for models that primarily depend on historical demand sequences.

Research comparing conventional forecasting approaches with more advanced methods has demonstrated the potential benefits of incorporating additional spatial and operational information. For example research on urban railway passenger volume prediction has compared ARIMA with neural network based approaches and found that deep learning models can provide improved prediction performance when spatial and operational information is incorporated into the forecasting process. [ScienceDirect](#)

The limitation of traditional statistical forecasting in this context is therefore not necessarily its inability to model temporal patterns but its limited ability to simultaneously represent a larger set of nonlinear relationships between passenger demand and railway operating conditions.

2.3.2 Machine Learning Based Demand Forecasting

Machine learning provides an alternative approach by allowing demand to be estimated from multiple explanatory variables. Features such as historical ticket sales route characteristics train capacity passenger class calendar information holidays and previous demand values can be incorporated into a single predictive model.

For railway applications this approach is particularly useful because passenger demand is influenced by both historical behaviour and contextual variables. Instead of modelling demand exclusively as a sequential time series the prediction problem can be formulated as a supervised learning problem where the target variable represents passenger demand and the input variables represent temporal operational and route related characteristics.

This approach is relevant to the proposed CeylonRail Intelligence component because the implemented forecasting pipeline combines historical demand with engineered temporal and operational features. The use of machine learning therefore provides a

practical mechanism for representing relationships that may not be adequately captured through conventional time series models alone.

2.3.3 Deep Learning for Railway Passenger Demand

Deep learning has received increasing attention in transportation demand forecasting because neural networks can learn nonlinear relationships from large historical datasets. Long Short Term Memory networks are particularly relevant because their recurrent structure allows information from previous observations to influence subsequent predictions.

Yang et al. developed a spatio temporal LSTM model for urban railway passenger volume forecasting using historical passenger volume together with station origin destination information and railway operation data. Their study compared the proposed approach with LSTM ARIMA and nonlinear autoregressive neural network models using actual railway data from the Beijing Metro Airport Line. The results indicated that incorporating spatial and temporal information improved prediction performance.

This finding supports the use of LSTM within the present research as an experimental forecasting approach. However the existing study focuses on urban rail passenger volume prediction while the present research considers railway demand across routes train types passenger classes and calendar conditions within the Sri Lankan context.

2.3.4 Spatio Temporal Demand Forecasting

A major development in transportation forecasting is the recognition that demand is not determined only by time. Passenger demand is also associated with geographical relationships between origins and destinations operational routes stations and network structure.

The Sp LSTM research demonstrates this concept by combining temporal passenger volume information with origin destination matrices and railway operation data. The model therefore considers both historical demand patterns and spatial relationships within the railway network.

This concept is important for the proposed research because railway demand can differ considerably between routes and passenger classes. A forecasting model that considers route related variables together with temporal characteristics can provide more operationally meaningful predictions than a model based only on total daily passenger counts.

The present research therefore adopts a broader spatio temporal perspective through the inclusion of route characteristics train information passenger class calendar variables and historical demand features. The objective is not only to predict future demand but also to generate predictions that can subsequently support operational capacity allocation.

2.3.5 Research on Railway Seat Allocation

Railway seat allocation has traditionally been studied as a capacity management and revenue management problem. Wang et al. investigated stochastic seat allocation for passenger rail transportation under customer choice. Their research formulated single stage and multi stage stochastic programming models to determine the number of seats allocated to different cabin classes and train services. The study also incorporated passenger choice behaviour and evaluated allocation policies using simulation experiments.

This research demonstrates that seat allocation can be formulated as an optimization problem rather than relying exclusively on predetermined allocation rules. It also highlights the importance of demand uncertainty because future passenger requests cannot be known with certainty when allocation decisions are made.

Other railway research has investigated joint pricing and seat allocation where capacity allocation decisions depend on origin destination demand and passenger response. Such approaches demonstrate the potential of mathematical optimization for improving railway capacity utilization and operational decision making.

However many existing studies are primarily concerned with revenue management and passenger choice. The main decision objective is therefore often associated with maximizing revenue rather than directly connecting machine learning based demand prediction with passenger service requirements.

2.3.6 Equity Aware Seat Allocation

Recent research has expanded railway seat allocation beyond revenue optimization by considering equity between different passenger groups and origin destination pairs. Xu et al. proposed a railway seat allocation framework that explicitly considers equity. Their study formulated a Mixed Integer Linear Programming model for deterministic demand and extended the approach using stochastic programming and distributionally robust optimization for uncertain demand. The results demonstrated that optimization can improve equity between different origin destination groups while considering demand uncertainty.

This research is particularly relevant to the proposed component because the present system also considers fairness as an important aspect of seat allocation. Instead of allocating all available capacity according to the highest predicted demand alone the proposed allocation mechanism applies minimum and maximum allocation constraints to prevent extreme allocation differences between passenger classes.

The difference is that the proposed research combines this allocation concept directly with machine learning based demand prediction. Therefore predicted demand becomes an input to the allocation process rather than relying only on predefined demand assumptions.

2.3.7 Integration of Forecasting and Seat Allocation

The reviewed studies demonstrate two relatively mature research areas. The first is passenger demand forecasting where statistical machine learning and deep learning models are used to estimate future passenger volumes. The second is railway seat allocation where mathematical optimization and stochastic modelling are used to determine how limited capacity should be distributed.

However these areas are frequently investigated independently. Demand forecasting research commonly evaluates prediction accuracy without examining how predictions influence subsequent operational decisions. Similarly seat allocation research often assumes demand information as an input rather than developing the forecasting mechanism that produces those demand estimates.

The proposed CeylonRail Intelligence component addresses this separation by establishing a sequential decision framework:

Historical railway data → Demand forecasting → Predicted passenger demand → Seat allocation optimization → Capacity utilization and passenger satisfaction

This integration creates a direct relationship between prediction accuracy and operational decision making. A forecasting model is therefore evaluated not only according to prediction error but also according to its usefulness for subsequent seat allocation.

2.3.8 Comparison of Related Research

Table 1: Comparison of Related Research

Research area	Main approach	Main contribution	Limitation in relation to this research
Traditional demand forecasting	Statistical time series models	Models historical demand patterns	Limited representation of multiple nonlinear operational factors
Urban railway passenger forecasting	Spatio temporal LSTM	Combines temporal passenger volume with spatial and operational information	Focuses mainly on urban railway passenger volume
Railway seat allocation	Stochastic programming	Allocates capacity under demand uncertainty and passenger choice	Primarily focuses on allocation rather than machine learning based demand prediction
Equity aware seat allocation	MILP stochastic programming and distributionally robust optimization	Incorporates fairness into railway capacity allocation	Does not directly establish the complete forecasting and allocation pipeline used in this research

Proposed research	Machine learning deep learning and integer optimization	Connects demand prediction with adaptive class wise seat allocation	Focused on the available Sri Lankan railway dataset and controlled experimental conditions
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The comparison indicates that existing research provides strong foundations for both passenger demand forecasting and railway seat allocation. Nevertheless, there remains an opportunity to integrate these approaches into a single operational framework designed around Sri Lankan railway demand characteristics.

2.3.9 Relevance to the Proposed Research

The reviewed research directly informs the design of the individual component. Research on spatio temporal passenger forecasting supports the inclusion of temporal and route related information in the demand prediction process. Research on LSTM provides the theoretical basis for evaluating deep learning based sequence modelling. Research on stochastic and mathematical seat allocation demonstrates that railway capacity can be optimized under demand constraints. Recent equity focused research further supports the inclusion of fairness considerations in capacity allocation.

Based on these findings, the proposed research combines machine learning based demand forecasting with integer optimization based seat allocation. The resulting approach is intended to move from prediction towards operational decision support by using predicted demand to determine how available railway capacity should be distributed across passenger classes.

2.4 Existing systems products and techniques

2.4 Existing Systems Products and Techniques

Existing railway information systems demonstrate how digital technologies can support passenger information, ticketing, railway operations and capacity management. However the level of intelligence provided by these systems varies considerably. Many operational railway systems are primarily designed to manage predefined schedules, ticket transactions and passenger information rather than

continuously predicting passenger demand and adapting seat allocation according to predicted demand.

2.4.1 Conventional Railway Management Systems

Traditional railway management systems generally operate around predefined timetables, train schedules, routes and available rolling stock. These systems provide the operational foundation required for railway services by maintaining information about trains, stations, routes and scheduled services.

Passenger ticketing systems additionally maintain information about bookings, passenger classes and available seats. Such information can provide an important source of historical demand data. However the existence of ticketing information does not necessarily mean that the data is automatically converted into predictive intelligence for future capacity planning.

In a conventional environment seat allocation is commonly determined through predefined capacity rules. Once a train and its available classes have been configured the allocation may remain relatively static even when passenger demand changes between weekdays, weekends, holidays or different periods of the year.

For the proposed CeylonRail Intelligence system this represents an important opportunity. Historical ticketing information can be transformed into structured demand features and used to support predictive capacity management rather than being used only for transaction processing.

2.4.2 Railway Passenger Information Systems

Passenger information systems are another important category of railway technology. These systems generally provide information such as train schedules, service availability, station information and journey details. Their primary objective is to improve passenger access to operational information.

Although passenger information is important for railway service quality these systems are generally focused on communicating existing operational information rather than predicting future passenger demand.

The proposed research extends this concept by using historical and contextual information to estimate future demand. The resulting predictions can then support an operational decision such as the distribution of available seats between passenger classes.

Therefore the proposed component can be viewed as an analytical layer that complements conventional passenger information and railway management functions.

2.4.3 Online Railway Ticketing Systems

Digital ticketing systems provide a valuable source of passenger transaction data. Each transaction can potentially contain information about travel date, route, train service, passenger class and number of tickets sold.

From a research perspective these records are valuable because repeated transactions can reveal temporal and spatial demand patterns. For example passenger demand may change according to the month, day of the week, holiday periods and passenger class.

The individual research component uses a dataset containing **55,302 records covering the period from 2020 to 2024**. The data was transformed into a structured forecasting dataset with tickets sold as the target variable. Additional variables included route information, train type, passenger class, railway line, day type, distance, capacity, holiday indicators and COVID period information.

This demonstrates how information normally associated with ticketing can be transformed into a predictive analytics resource. The forecasting component therefore builds upon the type of operational information that would naturally be available within a digital railway environment.

2.4.4 Machine Learning Demand Forecasting Techniques

Several machine learning techniques can be applied to passenger demand forecasting. Tree based models are particularly useful when demand depends on a mixture of numerical and categorical variables.

The implemented research evaluates Random Forest and XGBoost using the prepared railway dataset. Random Forest produced an MAE of **24.10**, an RMSE of **37.67** and a

MAPE of **13.12 percent**. XGBoost produced an MAE of **24.20**, an RMSE of **38.13** and a MAPE of **13.43 percent**.

Based on these primary model results Random Forest was selected as the production demand forecasting model for the implemented component.

The use of these techniques provides an alternative to relying exclusively on traditional time series forecasting. The models can incorporate multiple explanatory variables and engineered temporal features within the same prediction process.

2.4.5 LSTM Based Forecasting

Long Short Term Memory networks provide another technique for modelling sequential passenger demand. LSTM networks are designed to retain relevant information from previous observations and are therefore suitable for problems where historical demand influences future demand.

An LSTM experiment was implemented as part of the research using a **14 day lookback period**. The architecture included a 64 unit LSTM layer followed by dropout and a 32 unit LSTM layer with subsequent dense layers. The model used the Adam optimizer with mean squared error loss and was trained for 30 epochs with a batch size of 16.

The LSTM experiment was evaluated using the 2024 test period. Class level results demonstrated different forecasting behavior across passenger classes. The first class produced an MAE of 3.60 while the second class produced an MAE of 21.45 and the third class produced an MAE of 47.27.

These results illustrate an important characteristic of railway demand forecasting. Prediction performance can vary substantially between passenger classes because demand volume and behavioral patterns are not necessarily identical across classes.

2.4.6 Integer Optimization for Seat Allocation

Forecasting passenger demand alone does not directly determine how available seats should be distributed. An additional decision mechanism is therefore required to transform predicted demand into an allocation plan.

The implemented system uses **PuLP** to formulate the seat allocation problem as an integer optimization model. The allocation process considers predicted demand together with train capacity and class allocation constraints.

The implemented constraints include:

1. Total allocated seats must equal the available train capacity.
2. The allocation for each class must be at least 70 percent of its base class capacity.
3. The allocation for each class must not exceed 120 percent of its base class capacity.
4. Satisfied demand cannot exceed the allocated number of seats.
5. Satisfied demand cannot exceed predicted passenger demand.

The objective of optimization is to maximize total satisfied demand.

This approach provides a direct connection between the forecasting output and the operational capacity decision. Instead of treating demand prediction as an isolated analytical task the predicted values become decision variables within the allocation process.

2.4.7 Comparison with Existing Techniques

Existing railway technologies and research techniques can therefore be divided into several categories.

Table 2: Comparison with Existing Techniques

System or technique	Primary purpose	Strength	Limitation addressed by proposed research
Conventional timetable systems	Schedule railway services	Provides stable operational planning	Does not dynamically respond to predicted passenger demand
Digital ticketing systems	Manage passenger transactions	Generates valuable historical demand information	Transaction data is not necessarily transformed into adaptive allocation decisions

Passenger information systems	Provide service information	Improves passenger access to railway information	Primarily communicates existing information rather than predicting demand
Statistical forecasting	Predict future demand	Effective for established temporal patterns	Limited when many nonlinear contextual variables influence demand
Random Forest	Machine learning demand prediction	Handles multiple feature types and nonlinear relationships	Requires suitable historical and contextual features
XGBoost	Machine learning demand prediction	Provides strong predictive modelling capability	Prediction alone does not determine capacity allocation
LSTM	Sequential demand prediction	Captures temporal dependencies	Requires sequence preparation and can produce different performance across passenger classes
Integer optimization	Capacity allocation	Provides constraint based allocation	Requires reliable demand estimates as input
Proposed integrated component	Demand forecasting and adaptive seat allocation	Connects predicted demand with operational allocation	Developed and evaluated using the available Sri Lankan railway dataset

2.4.8 Position of the Proposed Component

The review of existing systems and techniques indicates that railway technology already provides many of the basic information resources required for intelligent capacity management. Ticketing systems can provide historical passenger information while railway management systems provide operational and capacity information. Machine learning and deep learning techniques can transform these data sources into

demand predictions while optimization techniques can transform predictions into allocation decisions.

The main distinction of the proposed component is the **integration of these stages into a single analytical workflow**. The research does not treat demand forecasting and seat allocation as independent problems. Instead predicted passenger demand is directly supplied to the allocation engine where capacity constraints and allocation limits are considered.

The resulting workflow can be represented as:

Railway data → Data preprocessing → Feature engineering → Demand forecasting → Predicted class demand → Integer optimization → Adaptive seat allocation

2.5 Gap synthesis and conceptual framework

The literature review demonstrates that railway passenger demand forecasting and railway seat allocation have been investigated using a range of statistical machine learning deep learning and optimization techniques. However these research areas are frequently treated as separate problems. Demand forecasting studies generally concentrate on improving prediction accuracy while seat allocation studies primarily focus on determining how available capacity should be distributed. This separation creates an opportunity to develop an integrated approach in which predicted passenger demand directly supports railway capacity allocation.

2.5.1 Identified Research Gaps

Several gaps were identified from the reviewed literature and existing railway technologies.

Gap 1: Limited integration between demand forecasting and seat allocation

Existing demand forecasting research generally evaluates how accurately future passenger demand can be predicted. However prediction accuracy alone does not explain how the predictions can support operational railway decisions. Similarly seat allocation research often assumes that demand information is already available.

The proposed research addresses this gap by connecting the forecasting stage directly to an optimization based seat allocation stage. The predicted demand generated by the forecasting model becomes an input to the allocation engine.

Gap 2: Limited focus on Sri Lankan railway demand characteristics

Many existing studies focus on railway systems in other countries or on urban metro systems. Their passenger behaviour operational structures and network characteristics may differ from those of Sri Lanka.

The proposed research uses Sri Lankan railway related data covering the period from 2020 to 2024. The dataset contains 55,302 records and includes route train type passenger class railway line calendar information capacity and demand related variables.

This provides a context specific basis for investigating railway passenger demand patterns in Sri Lanka.

Gap 3: Limited consideration of multiple demand influencing factors

Passenger demand is affected by several temporal and operational characteristics. A forecasting system based only on previous passenger volume may not adequately represent variations associated with passenger class route characteristics of holidays day types and seasonal patterns.

The proposed component therefore incorporates multiple feature categories including calendar features cyclical representations lag variables and rolling demand statistics together with railway operational characteristics.

Gap 4: Limited connection between prediction uncertainty and capacity decisions

A prediction can provide an estimate of future demand but an operational system must still determine how limited capacity should be distributed. If predicted demand differs substantially between passenger classes a purely static allocation may result in underutilized capacity in some classes while demand remains unsatisfied in others.

The proposed allocation mechanism uses predicted demand together with minimum and maximum allocation constraints. This enables the allocation process to respond to estimated demand while maintaining operational restrictions.

Gap 5: Limited consideration of fairness within adaptive allocation

Maximizing satisfied demand alone can potentially result in excessive allocation towards a single passenger class. Therefore an adaptive allocation system should maintain reasonable allocation boundaries.

The proposed system addresses this issue through class allocation constraints. Each class is required to receive at least 70 percent of its base capacity and cannot exceed 120 percent of its base capacity. These constraints provide a controlled allocation mechanism rather than allowing unrestricted redistribution of seats.

2.5.2 Research Gap Synthesis

The identified gaps can be summarized through three interconnected research requirements:

- 1. Accurate demand estimation**

The system must estimate future passenger demand using historical railway data and relevant temporal and operational features.

- 2. Demand aware capacity allocation**

The predicted demand must be converted into an operational allocation decision.

- 3. Controlled and fair allocation**

The allocation process must operate within predefined capacity boundaries while attempting to maximize satisfied passenger demand.

These requirements establish the central research direction of the individual component:

Develop an integrated demand forecasting and adaptive seat allocation framework that uses AI based passenger demand predictions to support constrained railway capacity allocation within the Sri Lankan context.

2.5.3 Conceptual Framework

The conceptual framework of the proposed research consists of four main stages: **data inputs, demand forecasting, adaptive allocation and evaluation.**

Stage 1: Data Inputs

The first stage receives historical railway information. The implemented dataset contains 55,302 records covering 2020 to 2024. Relevant information includes passenger demand railway route characteristics train type passenger class capacity calendar information holidays and other temporal indicators.

Stage 2: Demand Forecasting

The prepared data is transformed through preprocessing and feature engineering. Temporal features include year month day of week and cyclical representations. Historical demand is represented through lag features and rolling averages.

The forecasting stage evaluates machine learning models including Random Forest and XGBoost. An LSTM model is also implemented as a deep learning experiment to investigate sequential demand prediction.

Stage 3: Adaptive Seat Allocation

The predicted passenger demand is transferred to the seat allocation engine. The allocation problem is formulated using integer optimization through PuLP.

The allocation engine considers:

- Available train capacity
- Predicted passenger demand
- Passenger class
- Base class capacity
- Minimum allocation constraints
- Maximum allocation constraints
- Satisfied passenger demand

The objective is to maximize the total amount of predicted demand that can be satisfied within the available capacity and allocation constraints.

Stage 4: Evaluation

The final stage evaluates both prediction performance and allocation performance. Forecasting performance is measured using MAE RMSE and MAPE. Allocation performance is examined using seat utilization and the relationship between allocated capacity and predicted demand.

2.5.4 Conceptual Relationship Between Variables

The conceptual relationship can be expressed as:

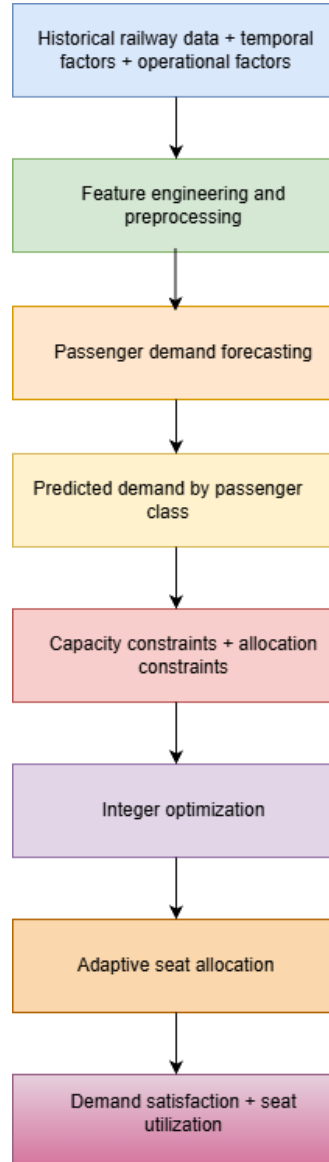


Figure 2.1 Conceptual Framework of the Proposed Demand Forecasting and Adaptive Seat Allocation Component

2.5.5 Conceptual Framework for the Individual Component

The conceptual framework can be represented as follows:

Table 3: Conceptual framework can be represented as

Input	Processing	Output
Historical ticket data	Data cleaning and preprocessing	Prepared railway dataset
Route information	Feature engineering	Route related features
Train information	Feature engineering	Operational features
Passenger class	Class based demand analysis	Class demand patterns
Calendar information	Temporal feature engineering	Temporal demand features
Holiday information	Seasonal analysis	Holiday demand indicators
Historical demand	Lag and rolling feature generation	Demand history features
Prepared features	Machine learning and deep learning	Predicted passenger demand
Predicted demand	Integer optimization	Adaptive seat allocation
Allocation results	Performance evaluation	Utilization and demand satisfaction measures

2.5.6 Overall Conceptual Model

The conceptual model establishes the individual research contribution as an integrated analytical pipeline rather than as an isolated forecasting model. The first part of the system answers the question "How much passenger demand is expected?" while the second part addresses "How should the available capacity be allocated according to that expected demand?"

This distinction is important because a useful railway intelligence system must transform analytical predictions into operationally meaningful decisions. The proposed component therefore combines predictive modelling with optimization to create a demand aware capacity management approach.

The framework also provides a basis for evaluating the practical value of the forecasting component. A forecasting model with lower prediction error is desirable but its operational usefulness can be further assessed by examining whether its predictions support effective seat utilization and demand satisfaction.

2.5.7 Summary of the Literature Review

The literature review established that statistical forecasting machine learning deep learning and optimization methods each provide valuable capabilities for railway demand management. Traditional forecasting approaches provide a foundation for modelling temporal behavior while machine learning can incorporate multiple explanatory variables. Deep learning models such as LSTM provide an alternative for modelling sequential patterns. Optimization techniques provide mechanisms for allocating limited railway capacity under operational constraints.

Despite these developments a clear opportunity remains to integrate passenger demand forecasting with adaptive seat allocation within a Sri Lankan railway context. The proposed research addresses this opportunity by developing a forecasting pipeline using historical railway data and connecting its predictions to an integer optimization based allocation engine.

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopts a quantitative experimental research design to investigate the effectiveness of artificial intelligence techniques for railway passenger demand forecasting and adaptive seat allocation. The research focuses on developing an integrated approach in which historical railway demand information is transformed into predictive information and subsequently used to support capacity allocation decisions.

The research design follows a structured sequence beginning with data collection and preparation followed by exploratory analysis feature engineering model development model evaluation and optimization based seat allocation. This structure enables the individual component to be evaluated at both the prediction level and the operational decision level.

The overall research process consists of the following major stages:

- 1. Data collection and preparation**
- 2. Exploratory data analysis**
- 3. Feature engineering**
- 4. Demand forecasting model development**
- 5. Forecasting model evaluation**
- 6. Adaptive seat allocation formulation**
- 7. Integer optimization**
- 8. Allocation evaluation**
- 9. Integration of forecasting and allocation results**

The research uses historical railway data covering the period from **2020 to 2024**. The available dataset contains **55,302 records**. The target variable used for demand forecasting is `tickets_sold`.

The dataset was divided temporally rather than randomly. Data from **2020 to 2023** was used for model training while data from **2024** was retained as the testing period. This approach is appropriate for a forecasting problem because future observations should not be used during model training. The temporal separation also provides a more realistic assessment of how the developed models perform when predicting demand for a later period.

3.1.1 Research Process

The research process begins with the preparation of historical railway demand information. The raw data is examined for inconsistencies and transformed into a structured format suitable for machine learning. Relevant railway operational variables and temporal characteristics are then incorporated into the forecasting dataset.

Feature engineering is performed to represent important demand patterns. These features include route information, train type, passenger class, railway line, day type, distance, capacity, holiday indicators, COVID period information and calendar variables. Historical demand information is additionally represented through lag and rolling statistical features.

Two primary machine learning models are evaluated during the forecasting stage:

- Random Forest
- XGBoost

An LSTM model is also implemented as a separate deep learning experiment to investigate sequential demand prediction.

The prediction results are then supplied to the adaptive seat allocation component. The allocation problem is formulated as an integer optimization problem using PuLP. The optimization model considers predicted demand and available capacity while maintaining predefined allocation constraints.

3.1.2 Forecasting Research Design

The forecasting component follows a supervised learning approach. The historical railway records are transformed into input features while `tickets_sold` is used as the prediction target.

The forecasting process can be expressed as:

Historical railway records → Feature engineering → Model training → Demand prediction → Model evaluation

The primary forecasting models are evaluated using Mean Absolute Error, Root Mean Square Error and Mean Absolute Percentage Error.

The Random Forest model achieved an MAE of **24.10**, an RMSE of **37.67** and a MAPE of **13.12 percent**. XGBoost achieved an MAE of **24.20**, an RMSE of **38.13** and a MAPE of **13.43 percent**. Based on these primary experimental results Random Forest was selected as the production forecasting model.

The LSTM experiment provides an additional investigation of sequential demand modelling. The implemented architecture uses a 14 day lookback period and was evaluated using the 2024 test period.

3.1.3 Adaptive Seat Allocation Research Design

The second part of the research transforms predicted passenger demand into an allocation decision. The allocation problem is formulated as an integer optimization model.

The model uses predicted passenger demand together with available train capacity and passenger class information. The implemented allocation constraints ensure that:

- Total allocated seats equal the available train capacity.
- Each passenger class receives at least 70 percent of its base capacity.
- Each passenger class receives no more than 120 percent of its base capacity.
- Satisfied demand cannot exceed the allocated seats.
- Satisfied demand cannot exceed predicted demand.

The objective of the optimization model is to maximize total satisfied passenger demand while maintaining the specified allocation constraints.

The resulting research design therefore connects the two main stages:

Demand forecasting → Predicted demand → Integer optimization → Adaptive seat allocation

3.1.4 Experimental Evaluation

The research evaluates the forecasting and allocation components separately before considering their combined operational relationship.

Forecasting performance is evaluated using quantitative prediction metrics. Allocation performance is examined through seat utilization and the relationship between predicted demand and allocated capacity. Class wise results are also considered to determine whether the allocation mechanism distributes capacity within the predefined boundaries.

This evaluation approach allows the research to answer two related questions. First it determines whether the developed models can provide useful passenger demand predictions. Second it investigates whether those predictions can be converted into feasible adaptive seat allocation decisions.

3.1.5 Alignment with Research Objectives

The research design directly supports the approved objectives of the individual component.

Table 4: Research Objective and Methodological approach

Research objective	Methodological approach
Collect and preprocess railway demand information	Dataset preparation and preprocessing
Construct demand related features	Temporal feature engineering, lag features and rolling features
Develop demand forecasting models	Random Forest, XGBoost and LSTM
Formulate adaptive seat allocation	Integer optimization using PuLP

Apply capacity and allocation constraints	Minimum and maximum class allocation constraints
Evaluate forecasting performance	MAE, RMSE and MAPE
Evaluate allocation performance	Seat utilization and demand satisfaction
Examine adaptive allocation	Comparison of predicted demand with allocated capacity

3.2 Data participants or study materials

The research is based on historical railway demand data and the software artefacts developed for the demand forecasting and adaptive seat allocation component. The primary study material is the railway ticketing dataset used to train and evaluate the demand forecasting models. Additional study materials include the trained machine learning models, feature encoders, forecasting outputs, allocation results and evaluation files generated during the experiments.

3.2.1 Primary Dataset

The primary dataset used in the research is **sl_railway_ticketing_2020_2024.csv**. The dataset contains **55,302 records** covering the period from **1 January 2020 to 31 December 2024**. The data contains three passenger class records for each train service pattern.

The main variables available in the dataset include:

Table 5: Main variables available

Variable	Description
date	Date associated with the railway service
year	Year of the observation
month	Month of the observation
day_of_week	Day of the week
day_type	Type of day

is_public_holiday	Indicates whether the date is a public holiday
holiday_name	Name of the relevant holiday
is_covid_lockdown_period	Indicates the COVID related lockdown period
origin_station	Origin station
destination_station	Destination station
line_name	Railway line
distance_km	Route distance
train_type	Type of train
class	Passenger travel class
seat_capacity	Available seat capacity
tickets_sold	Number of tickets sold
seat_utilization_pct	Recorded seat utilization
fare_lkr	Ticket fare in Sri Lankan Rupees
revenue_lkr	Recorded revenue
delay_minutes	Recorded delay duration

3.2.2 Training and Testing Data

A chronological data separation was used to maintain the temporal structure of the forecasting problem. Records before **1 January 2024** were used for model training while records from **1 January 2024 onwards** were used for testing. The resulting dataset contained **44,928 training records** and **10,374 testing records**.

This approach avoids using future observations during model training. It also provides a more realistic evaluation because the model is trained using earlier railway demand patterns and subsequently evaluated using a later period.

The chronological division can therefore be represented as:

2020 to 2023 → Training data

2024 → Testing data

3.2.3 Feature Variables

The forecasting implementation uses a combination of categorical numerical temporal and historical demand features.

Categorical variables include route train type passenger class railway line and day type. These variables are converted into numerical representations using label encoding. The trained encoders are stored separately so that the same representations can be used when the trained model is applied to new prediction requests.

Temporal features include month day of the weekday week of the year weekend status and cyclical representations of month and day of the week. These features allow the model to represent recurring temporal patterns within passenger demand.

Historical demand features include:

- lag_1
- lag_7
- roll_mean_7
- roll_mean_30

These features represent recent demand behaviour and longer recent demand trends for the relevant route train type and passenger class. Pasted text(20261004-120308)

The implementation also contains environmental features in the latest model training workflow. These include average temperature total route precipitation total route rain total route precipitation hours and average route wind speed. The training metadata identifies Open Meteo ERA5 archive data as the historical weather source.

3.2.4 Data Preparation for the Forecasting Model

The dataset is transformed before being provided to the forecasting algorithms. Categorical variables are encoded while numerical and temporal variables are retained or derived as required by the model.

The route representation is created from the origin and destination information. Railway line train type passenger class and day type are also encoded. Temporal

variables are derived from the service date while historical demand variables are calculated using previous observations.

The resulting feature representation enables the forecasting model to consider both the characteristics of the railway service and the historical behaviour of passenger demand.

3.2.5 Study Materials and Research Artefacts

In addition to the primary dataset the research generated several artefacts during model development and evaluation. These include:

- actual_vs_predicted.csv
- class_wise_metrics.csv
- class_wise_allocation_summary.csv
- dashboard_summary.json
- demand_by_class_timeseries.csv
- peak_time_by_dayofweek.csv
- seasonal_factors_by_month.csv
- seasonal_holiday_summary.json
- seat_allocation_results.csv
- Demand analysis charts
- demand_forecast_model.joblib
- label_encoders.joblib
- lstm_aggregate_demand.keras

These artefacts provide supporting evidence for the forecasting experiments analysis and allocation results.

3.2.6 Forecasting Models

The primary production forecasting model is a **Random Forest Regressor**. The implemented configuration uses 300 estimators with a maximum tree depth of 14 and a fixed random state of 42. XGBoost was also trained and evaluated during the

research. The model with the lowest MAE was selected as the production demand forecasting model and the resulting model was Random Forest.

An LSTM model was additionally implemented to investigate sequential demand forecasting. The LSTM uses a 14 day lookback period with two LSTM layers dropout and dense layers. The experiment uses the 2024 period as the chronological test period.

It is important to distinguish the LSTM experiment from the production forecasting model. LSTM produces an offline forecasting artefact and is not loaded by the current live demand prediction service. The current live service uses the saved Random Forest model.

3.2.7 Seat Allocation Study Materials

The predicted demand generated by the forecasting process is used as an input for the adaptive seat allocation component. The allocation results are stored through research artefacts including `seat_allocation_results.csv` and `class_wise_allocation_summary.csv`.

The allocation stage therefore uses the forecasting output as the demand input while considering railway seat capacity and passenger class requirements. This allows the research to evaluate the transition from passenger demand prediction to an operational seat allocation decision.

3.2.8 Dataset Limitation

The available project materials establish the structure size and time period of the dataset but they do not establish whether the complete `sl_railway_ticketing_2020_2024.csv` dataset consists entirely of real operational Sri Lankan Railway records generated records or a combination of sources. Therefore this thesis does not make an unsupported claim about the complete provenance of the dataset.

This limitation is important when interpreting the experimental findings and is considered further in the limitations and threats to validity section.

3.2.9 Summary

The study uses a structured railway demand dataset containing 55,302 records from 2020 to 2024. A chronological division separates 2020 to 2023 for training from 2024 for testing. The forecasting process uses railway service characteristics temporal information historical demand features and additional environmental features in the latest model workflow. Random Forest is used as the production forecasting model while LSTM provides a separate sequential forecasting experiment. The resulting demand predictions are subsequently used to support adaptive seat allocation.

3.3 Experimental or study procedure

The experimental procedure was designed to develop and evaluate the demand forecasting and adaptive seat allocation components of the CeylonRail Intelligence system. The procedure follows a sequence of data preparation, exploratory analysis, feature engineering, model development, forecasting evaluation and optimization based seat allocation.

The experiments use the historical railway dataset covering 2020 to 2024. The forecasting models are trained using earlier observations and evaluated using the 2024 testing period. The resulting predictions are then used by the seat allocation component to determine how available capacity can be distributed according to estimated passenger demand.

3.3.1 Step 1: Data Collection and Preparation

The first step involved loading the railway ticketing dataset into the Python development environment. The dataset contains 55,302 records with information about railway services, passenger classes, ticket sales, dates, routes and operational characteristics.

The target variable was defined as `tickets_sold`, representing the observed demand for the relevant railway service and passenger class.

The dataset was examined to identify the variables required for forecasting and to prepare the data for subsequent analysis. The preparation process focused on maintaining the relationship between dates, railway routes, train types and passenger classes.

The chronological split was then applied. Records dated before 1 January 2024 were assigned to the training dataset while records dated from 1 January 2024 onwards were assigned to the testing dataset. This resulted in 44,928 training records and 10,374 testing records.

3.3.2 Step 2: Exploratory Data Analysis

Exploratory data analysis was conducted to understand the distribution of passenger demand and identify patterns that could influence forecasting and capacity allocation.

The analysis examined demand across passenger classes and months as well as differences between weekdays and weekends. Holiday related demand patterns and variations across the available railway services were also considered.

Visualizations were generated to examine:

- Total demand by passenger class
- Average demand by month
- Daily demand trends
- Demand patterns across days of the week
- Average seat utilization during holidays and non holiday periods
- Seasonal demand factors

These visualizations provided a descriptive understanding of the dataset and supported the identification of temporal and operational characteristics relevant to model development.

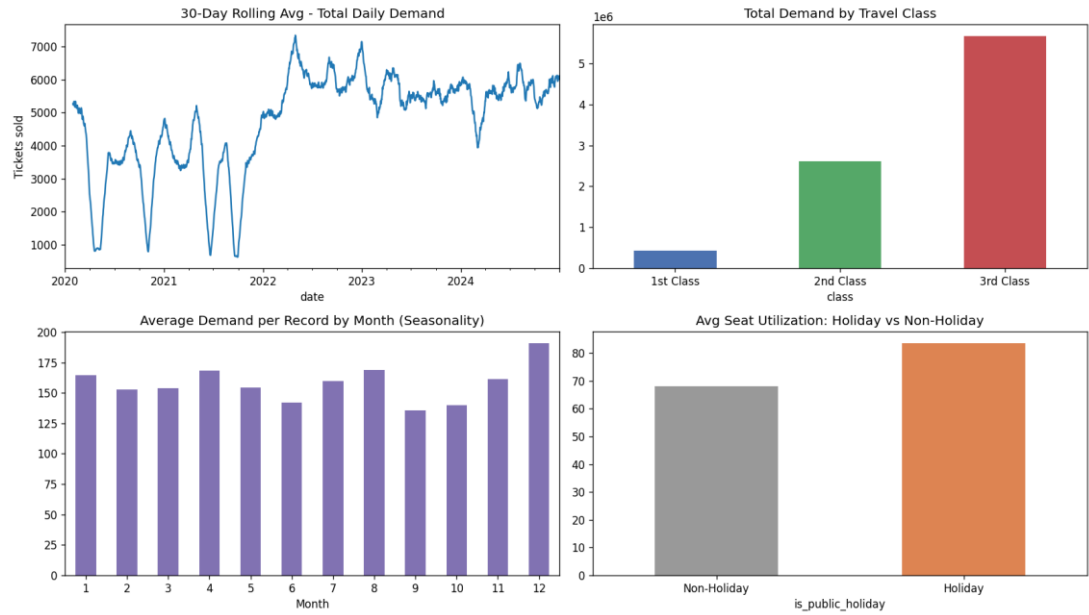


Figure 3: Exploratory Data Analysis of Railway Passenger Demand

3.3.3 Step 3: Feature Engineering

Feature engineering was performed to represent the characteristics of railway services and the temporal behaviour of passenger demand.

The categorical variables included route, train type, passenger class, railway line and day type. These variables were converted into numerical representations using label encoding.

Temporal features were derived from the service date. These included month, day of the week, day of the month, week of the year and weekend status. Cyclical encoding was also applied to month and day of the week to represent recurring temporal patterns.

Historical demand features were constructed to represent previous demand observations. These included lag features and rolling averages.

The main engineered features included:

Purpose

- Represents the railway route
- Represents the train service category
- Represents the passenger class
- Represents the railway line

- Represents the type of day
- Represents route distance in kilometres
- Represents the available class capacity
- Represents public holiday conditions
- Represent temporal characteristics
- Represent recurring calendar patterns
- Represents the previous available demand observation
- Seven day lag
- Represents demand from a previous weekly observation
- Represents recent average demand
- Thirty observation rolling mean
- Represents a longer recent demand trend

The historical demand features were calculated using previous records associated with the relevant route, train type and passenger class. This allowed the forecasting models to consider both recent observations and broader demand patterns.

The latest model training workflow also incorporates environmental variables including temperature, precipitation, rainfall duration and wind speed. These variables provide an additional representation of weather conditions that may be associated with passenger demand.

3.3.4 Step 4: Training the Random Forest Model

Random Forest was implemented as the primary production forecasting model. The model was trained using the prepared feature dataset and the `tickets_sold` target variable.

The implemented configuration used 300 estimators, a maximum tree depth of 14 and a fixed random state of 42. The model was trained using the historical training period and subsequently evaluated using the 2024 test dataset.

Random Forest was selected as the production model because it achieved lower prediction error than XGBoost in the primary model comparison.

The trained model and the associated categorical encoders were saved for reuse by the demand prediction service. This allowed the forecasting model developed during experimentation to be used by the wider CeylonRail Intelligence application.

3.3.5 Step 5: Training and Evaluating XGBoost

XGBoost was implemented as a second primary forecasting model. It was trained using the prepared railway demand features and evaluated against the same chronological testing period.

The comparison was intended to determine whether XGBoost provided a more accurate alternative to Random Forest for the available dataset.

The primary comparison produced the following results:

Table 6:primary comparison produced results

Model	MAE	RMSE	MAPE
Random Forest	24.10	37.67	13.12%
XGBoost	24.20	38.13	13.43

Random Forest achieved lower values for all three reported error metrics. It was therefore selected as the production forecasting model for the implemented system.

3.3.6 Step 6: LSTM Experiment

An additional experiment was conducted using a Long Short Term Memory network to investigate sequential demand forecasting.

The model used a 14 day lookback period to represent previous demand observations. The implemented architecture consisted of a 64 unit LSTM layer, a dropout layer with a rate of 0.2, a 32 unit LSTM layer and subsequent dense layers.

The model used the Adam optimizer with mean squared error loss. Training was performed for 30 epochs using a batch size of 16.

The experiment used the 2024 period for testing. Actual and predicted demand values were compared to examine the model's forecasting behaviour. Class wise evaluation was also performed.

The recorded class wise results were:

Table 7: Recorded class wise results

Passenger Class	MAE	RMSE	MAPE
First Class	3.60	4.51	13.11%
Second Class	21.45	26.82	13.09%
Third Class	47.27	59.32	13.15%

These results show that the absolute prediction errors differed between passenger classes. The values should be interpreted in relation to the demand scale of each class.

The LSTM was retained as a separate experimental model. The current live demand prediction service uses the saved Random Forest model rather than the LSTM model.

3.3.7 Step 7: Formulating the Seat Allocation Problem

After the forecasting stage, the predicted passenger demand was used as an input to the adaptive seat allocation component.

The allocation problem was formulated as an integer optimization model using the PuLP library. The purpose was to distribute available train capacity across passenger classes while respecting the predefined allocation boundaries.

The optimization process considered predicted demand, total train capacity and the base capacity assigned to each passenger class.

The model incorporated the following constraints:

1. The total number of allocated seats must equal the available train capacity.
2. Each class must receive at least 70 percent of its base capacity.
3. Each class must receive no more than 120 percent of its base capacity.
4. Satisfied demand cannot exceed allocated seats.
5. Satisfied demand cannot exceed predicted demand.

The objective was to maximize the total satisfied passenger demand within these constraints.

This formulation enables predicted demand to influence allocation decisions while preserving the specified capacity requirements.

3.3.8 Step 8: Generating Allocation Results

The optimization model was used to generate seat allocation results based on the predicted demand and capacity constraints.

The resulting outputs were organized for subsequent analysis. These included class wise allocation summaries, seat allocation results and visualizations comparing predicted demand with allocated capacity.

The analysis examined whether the available capacity was distributed according to predicted demand and whether the resulting allocation respected the configured class boundaries.

The allocation results were also used to examine seat utilization across passenger classes. This provided an operational perspective on the forecasting component by showing how demand estimates could support capacity allocation decisions.

3.3.9 Step 9: Evaluating the Integrated Component

The final experimental stage considered the forecasting and allocation components together.

Forecasting performance was evaluated using MAE, RMSE and MAPE. Allocation behavior was examined using seat utilization and the relationship between predicted demand and allocated capacity.

The analysis also considered differences between passenger classes and temporal demand patterns. These results were used to assess the extent to which the forecasting output could support the intended seat allocation process.

The evaluation was based on the implemented experiments and available research artefacts. No unsupported performance improvement was ensured for the integrated system. In particular any claim that adaptive allocation outperforms a static allocation

baseline would require a clearly defined comparative experiment and corresponding measured results.

3.4 Measures and evaluation metrics

This research uses quantitative evaluation measures to assess the accuracy of passenger demand forecasting and examine the performance of the adaptive seat allocation component. The evaluation is divided into two main areas: forecasting performance and allocation performance.

Forecasting performance is evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE). Allocation performance is examined through seat utilization, the relationship between predicted demand and allocated capacity and the amount of demand that can be satisfied within the specified constraints.

3.4.1 Mean Absolute Error

Mean Absolute Error measures the average absolute difference between actual passenger demand and predicted passenger demand. It provides an indication of how far the model predictions deviate from the observed values.

The formula is:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

A lower MAE indicates that the model predictions are closer to the actual demand values on average. Since the target variable represents tickets sold, MAE is expressed in the same units as the target.

MAE is particularly useful for this research because it provides a direct interpretation of the average absolute prediction error.

3.4.2 Root Mean Square Error

Root Mean Square Error measures the square root of the average squared difference between actual and predicted demand.

The formula is:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE assigns greater weight to larger prediction errors because the differences are squared before averaging. This makes the metric useful for identifying models that occasionally produce substantial demand prediction errors.

A lower RMSE indicates better predictive performance under this measure. Like MAE, RMSE is expressed in the units of the target variable.

3.4.3 Mean Absolute Percentage Error

Mean Absolute Percentage Error measures prediction error relative to actual demand and expresses the result as a percentage.

The formula is:

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

MAPE allows the forecasting performance of different models or passenger classes to be examined on a relative scale.

However, MAPE becomes undefined when actual demand is zero and can become disproportionately large when actual demand is close to zero. The implementation must therefore handle zero demand values explicitly. The results should be interpreted with this limitation in mind.

3.4.4 Comparison of Forecasting Models

The primary comparison uses the MAE, RMSE and MAPE values obtained from the Random Forest and XGBoost experiments.

Table 8: Random Forest and XGBoost experiments.

Evaluation metric	Random Forest	XGBoost	Preferred result
-------------------	---------------	---------	------------------

MAE	24.10	24.20	Random Forest
RMSE	37.67	38.13	Random Forest
MAPE	13.12%	13.43	Random Forest

Random Forest achieved lower error values across all three reported metrics. It was therefore selected as the production demand forecasting model.

The difference between the two models was relatively small. Consequently, the results establish that Random Forest performed better in the reported comparison but do not independently demonstrate that it will outperform XGBoost across all railway routes, passenger classes or future periods.

3.4.5 Limitations of the Evaluation Measures

The selected measures provide complementary information about forecasting accuracy and allocation behaviour. Nevertheless, several limitations must be considered.

First, MAE and RMSE are expressed in the units of the target variable. Their values can therefore be difficult to compare directly between passenger classes with substantially different demand scales.

Second, MAPE can be problematic when actual demand is zero or close to zero. Any reported MAPE value must be interpreted according to the method used to handle such observations.

Third, predicted seat utilization does not necessarily equal actual occupancy. Actual utilization requires reliable observations of the number of passengers who occupied the available seats.

Finally, allocation feasibility does not by itself prove that the proposed allocation method performs better than a static allocation policy. Such a conclusion requires a defined baseline and a controlled comparison using consistent demand inputs and capacity conditions.

3.5 Data analysis

Data analysis was conducted to evaluate the forecasting performance of the developed models and examine how predicted passenger demand can support adaptive seat

allocation. The analysis used the prepared railway dataset covering 2020 to 2024 and the experimental outputs generated during model development.

The forecasting results were analysed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE). Random Forest achieved an MAE of 24.10, an RMSE of 37.67 and a MAPE of 13.12 percent. XGBoost achieved an MAE of 24.20, an RMSE of 38.13 and a MAPE of 13.43 percent. These results indicate that Random Forest performed better across the three reported metrics and was selected as the production forecasting model.

Exploratory analysis was used to examine passenger demand patterns across travel classes, months, days of the week and holiday periods. Visualizations of daily demand trends, monthly demand patterns and class wise seat utilization helped identify variations in passenger demand and provided context for interpreting the forecasting results.

The LSTM experiment was analysed separately to investigate sequential demand prediction. Its class wise results were examined using MAE, RMSE and MAPE. The results showed different absolute prediction errors across passenger classes while the reported MAPE values remained similar.

For seat allocation, the analysis examined predicted demand, allocated capacity, satisfied demand and seat utilization across passenger classes. The allocation results were assessed against the capacity constraints implemented in the PuLP optimization model. This analysis helped determine whether the generated allocations complied with the configured limits and how the allocation process responded to predicted demand.

The findings from these analyses were used to assess the forecasting component and the technical feasibility of integrating demand predictions with seat allocation. However, the available results alone do not establish that adaptive allocation outperforms static allocation. Such a conclusion requires a documented comparative experiment using a defined static allocation baseline.

3.6 Ethics privacy security and responsible AI

This research uses railway ticketing data to develop passenger demand forecasting and adaptive seat allocation. The analysis focuses on aggregated demand patterns and railway service characteristics rather than identifying individual passengers.

The dataset includes variables such as travel dates, routes, passenger classes, seat capacity and tickets sold. Any personal or identifiable passenger information should be excluded from the research dataset. The available project materials do not establish whether the original data contains personal information or whether formal ethical approval was obtained. These details should therefore be confirmed before the thesis is submitted.

Responsible use of the forecasting model requires careful interpretation of its predictions. Forecasting errors may affect the estimated demand for individual passenger classes. Therefore, predictions should be treated as estimates rather than guaranteed passenger numbers.

The allocation component applies minimum and maximum class capacity constraints to control the redistribution of seats. These constraints support a more controlled allocation process, although they do not independently prove fairness between all passenger groups.

The research also acknowledges the limitations of the available dataset and the possibility that future demand may differ from historical patterns. These considerations are important when interpreting experimental results and assessing the suitability of the proposed component for real railway operations.

3.7 Reproducibility and research artefacts

The research maintains reproducibility through the use of a defined dataset, a chronological training and testing split and documented model configurations. The dataset contains 55,302 records covering 2020 to 2024, with records from 2020 to 2023 used for training and records from 2024 used for testing.

The primary forecasting model is a Random Forest Regressor configured with 300 estimators, a maximum depth of 14 and a fixed random state of 42. The trained model

and categorical encoders are saved as `demand_forecast_model.joblib` and `label_encoders.joblib`. The separate LSTM experiment is saved as `lstm_aggregate_demand.keras`.

Additional research artefacts include forecasting metrics, actual and predicted demand values, class wise allocation summaries, seat allocation results and demand analysis visualizations. These outputs provide supporting evidence for the experiments and assist with examining the results presented in Chapter 6.

The seat allocation component uses the PuLP optimization library to formulate the capacity allocation problem. Reproducing the allocation results requires the same input predictions, base capacities, total train capacity, allocation constraints and optimization objective.

For reliable reproduction, the dataset version, feature definitions, model configurations, software dependencies and evaluation procedures should be retained with the research materials. The reported results should be regenerated from the saved artefacts or original experiments whenever possible rather than relying solely on manually recorded values.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The demand forecasting and adaptive seat allocation component of CeylonRail Intelligence is designed to predict railway passenger demand and use the predictions to support seat allocation decisions. The component combines historical railway data, machine learning models and integer optimization to provide demand based allocation within defined capacity constraints.

The main functional requirements are:

1. Data processing: The system must process railway demand data containing travel dates, routes, train types, passenger classes, seat capacity and historical ticket sales.
2. Demand forecasting: The system must predict passenger demand using the trained Random Forest model. The LSTM model is used separately for experimental evaluation.
3. Temporal analysis: The system must support analysis of demand variations across dates, months, weekdays, weekends and holiday periods.
4. Class wise prediction: The system must produce demand predictions for the relevant passenger classes.
5. Adaptive seat allocation: The system must use predicted demand to determine seat allocation across passenger classes.
6. Capacity constraints: The allocation process must respect total train capacity and the defined minimum and maximum allocation limits.
7. Performance evaluation: The system must support forecasting evaluation using MAE, RMSE and MAPE and analysis of allocation results using seat utilization and demand satisfaction.

The main non functional requirements are:

- Accuracy: Forecasting performance must be evaluated using established quantitative metrics.

- **Reliability:** The system must use consistent feature preparation and saved model encoders when generating predictions.
- **Performance:** Demand prediction and allocation should be completed within a practical time for dashboard use.
- **Maintainability:** The forecasting and allocation components should remain sufficiently modular to support future updates.
- **Usability:** Results should be presented clearly through demand charts, class wise comparisons and allocation summaries.
- **Constraint compliance:** Generated allocations must satisfy the capacity and allocation constraints defined in the optimization model.

These requirements establish the expected behaviour of the individual component and provide the basis for the architecture and detailed design described in the following sections.

4.2 Component architecture within the complete solution

The demand forecasting and adaptive seat allocation component is part of the wider CeylonRail Intelligence system. The overall system includes four main components: passenger demand forecasting and seat allocation, train tracking and delay prediction, AI fraud detection and passenger verification and scheduling optimization.

Within this architecture, the individual component processes railway demand information, generates passenger demand predictions and uses the predicted values to support seat allocation decisions.

The component follows a modular architecture consisting of a data layer, forecasting layer, optimization layer and presentation layer.

The data layer stores the historical railway dataset and provides the information required for model training, prediction and analysis. Relevant data includes travel dates, routes, train types, passenger classes, seat capacity and ticket sales.

The forecasting layer processes the input features and generates demand predictions using the trained Random Forest model. The separate LSTM model supports experimental evaluation of sequential demand forecasting.

The optimization layer receives the predicted demand and determines seat allocation using an integer optimization model implemented with PuLP. The model applies total capacity constraints and minimum and maximum allocation limits for each passenger class.

The presentation layer displays demand forecasts, historical demand patterns, class wise comparisons and allocation results through the system dashboard. This allows railway staff to interpret the predictions and examine the proposed allocation.

The component operates through the following process:

Railway data → Feature preparation → Random Forest demand prediction → Predicted demand by class → Integer optimization → Adaptive seat allocation → Dashboard presentation

The architecture separates prediction from allocation so that each stage can be evaluated independently. This also supports future improvements to the forecasting model or allocation algorithm without requiring the entire system to be redesigned.

The component communicates with the wider CeylonRail Intelligence system through its backend services and shared railway data resources. Its main contribution is to transform historical railway demand information into predictions and use those predictions to support capacity management.

4.3 Design alternatives and decisions

Several approaches were considered for implementing passenger demand forecasting and adaptive seat allocation. The final design uses Random Forest for production demand prediction and integer optimization for seat allocation. An LSTM model was also implemented as a separate experiment to investigate sequential demand forecasting.

For demand forecasting, Random Forest and XGBoost were evaluated using the same chronological training and testing approach. Random Forest achieved an MAE of

24.10, an RMSE of 37.67 and a MAPE of 13.12 percent, while XGBoost achieved an MAE of 24.20, an RMSE of 38.13 and a MAPE of 13.43 percent. Random Forest was selected because it produced lower errors across all three reported metrics.

LSTM was selected for an additional experiment because its recurrent architecture can model sequential demand patterns. The experiment used a 14 day lookback period and evaluated predictions using the 2024 testing period. However, the LSTM remains separate from the production forecasting service, which uses Random Forest.

For seat allocation, integer optimization was selected instead of relying only on fixed allocation rules. The implementation uses PuLP to maximize satisfied predicted demand while respecting total train capacity and the minimum and maximum allocation limits defined for each passenger class.

The principal design decisions are summarized below.

Table 9: Design decisions summarized

Design area	Selected approach	Reason
Production forecasting	Random Forest	Achieved the lowest reported MAE, RMSE and MAPE in the primary model comparison
Additional forecasting experiment	LSTM	Investigates sequential demand patterns
Categorical feature preparation	Label encoding	Provides numerical representations for categorical model inputs
Historical demand representation	Lag features and rolling averages	Represents recent demand behaviour
Seat allocation	Integer optimization using PuLP	Supports demand based allocation under defined constraints
Evaluation	Forecasting error metrics and allocation analysis	Assesses predictive accuracy and allocation behaviour

The selected design connects passenger demand prediction with operational capacity allocation while keeping the forecasting and optimization stages separate. This structure supports independent evaluation of each stage and provides a foundation for future improvements to the individual component.

4.4 Detailed design

The detailed design of the proposed component consists of two main processes: passenger demand forecasting and adaptive seat allocation. The forecasting process estimates passenger demand for each relevant railway service and passenger class while the allocation process uses the predicted demand to distribute available seats within predefined capacity constraints.

The forecasting process begins by extracting relevant features from the prepared railway dataset. These include route, train type, passenger class, railway line, date related variables, seat capacity, holiday indicators and historical demand features. Categorical variables are encoded using the saved label encoders and temporal features are generated consistently with the training process.

The trained Random Forest model receives the prepared features and predicts the expected number of tickets sold. These predictions are then passed to the allocation process. An LSTM model is also available for offline experimental analysis but is not used by the current live demand prediction service.

The allocation process is implemented using PuLP. It uses predicted demand and available train capacity to determine the allocation across passenger classes. The model aims to maximize total satisfied demand while ensuring that the total allocation equals available capacity and that each class remains within the specified minimum and maximum allocation limits.

The final outputs include predicted passenger demand, class wise seat allocation and seat utilization information. These results can be presented through the CeylonRail Intelligence dashboard to support the interpretation of demand patterns and capacity allocation decisions.

The separation of forecasting and optimization allows each process to be evaluated independently while maintaining a clear connection between the predicted demand and the resulting allocation.

4.5 Standards and professional practice

The development of the demand forecasting and adaptive seat allocation component follows established software engineering and data science practices. The implementation separates data preparation, forecasting and optimization to improve maintainability and support independent evaluation of each stage.

The forecasting models are evaluated using MAE, RMSE and MAPE to provide measurable evidence of prediction performance. A chronological training and testing split is used to evaluate predictions against a later period. The trained model and categorical encoders are saved to support consistent feature processing during subsequent predictions.

The seat allocation component applies explicit capacity constraints through integer optimization. This ensures that allocation decisions are evaluated against the configured operational limits rather than being determined solely by predicted demand.

Professional practice also requires transparent reporting of experimental results and limitations. The thesis distinguishes the production Random Forest model from the separate LSTM experiment and avoids claiming that adaptive allocation outperforms static allocation without a documented comparative evaluation.

These practices support reproducibility, maintainability and responsible interpretation of the research findings.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The demand forecasting and adaptive seat allocation component of CeylonRail Intelligence was developed using Python and machine learning libraries. The implementation covers railway data preparation, feature engineering, demand forecasting, model evaluation and optimization based seat allocation.

The primary dataset contains 55,302 records covering the period from 2020 to 2024. Random Forest was selected as the production forecasting model following comparison with XGBoost. An LSTM model was implemented separately to investigate sequential demand forecasting. The seat allocation component uses the PuLP library to formulate an optimization problem based on predicted demand and capacity constraints.

Table 10: Development Technologies and Their Purposes

Technology	Application in the Research
Python	Implementation of data processing, forecasting and seat allocation
Pandas	Loading, transforming and analysing railway demand data
NumPy	Numerical calculations and feature preparation
Scikit learn	Random Forest implementation, categorical encoding and evaluation metrics
XGBoost	Implementation of the additional forecasting model for comparison
TensorFlow and Keras	Implementation of the LSTM forecasting experiment

PuLP	Formulation and solution of the integer optimization problem for seat allocation
Joblib	Saving and loading the trained Random Forest model and categorical encoders
Matplotlib	Generation of demand analysis and forecasting visualizations
CSV files	Storage of railway data, predictions and experimental results
JSON files	Storage of structured analytical outputs and dashboard summaries

The production forecasting model and its categorical encoders are saved as `demand_forecast_model.joblib` and `label_encoders.joblib`. The separate LSTM experiment is saved as `lstm_aggregate_demand.keras`. These artefacts support the reuse of trained models and the examination of experimental results.

The forecasting process generates demand predictions for individual passenger classes. These predictions are used by the allocation component to determine seat distribution while respecting the defined train capacity and class allocation constraints. The resulting outputs support demand analysis and the presentation of allocation results through the wider CeylonRail Intelligence system.

5.2 Component implementation

The individual component was implemented as two connected processes: passenger demand forecasting and adaptive seat allocation. Python was used for data processing, model development and optimization. The production forecasting service uses a trained Random Forest model while the LSTM model was developed separately for experimental analysis.

The forecasting process uses historical railway data containing 55,302 records from 2020 to 2024. Feature preparation includes categorical encoding, calendar variables, cyclical date features and historical demand features such as lag values and rolling averages. The target variable is `tickets_sold`.

The Random Forest model was configured with 300 estimators and a maximum depth of 14. XGBoost was also evaluated for comparison. Random Forest was selected for production because it achieved lower MAE, RMSE and MAPE values in the recorded comparison. The trained model and categorical encoders were saved using Joblib for reuse by the demand prediction service.

A separate LSTM experiment was implemented to examine sequential demand patterns using a 14 day lookback period. Its evaluation results were recorded separately from the production model results. The LSTM experiment is not used by the current live demand prediction service.

The seat allocation component was implemented using PuLP. It receives predicted demand and applies capacity constraints to determine the allocation across passenger classes. The optimization objective is to maximize satisfied demand while maintaining the specified total capacity and class allocation limits. The generated outputs include class wise allocation results and seat utilization information.

The forecasting and allocation processes are designed as separate stages. This separation allows forecasting accuracy and allocation feasibility to be examined independently while enabling predicted demand to inform allocation decisions.

5.3 Integration with the other components

The demand forecasting and adaptive seat allocation component forms part of the CeylonRail Intelligence system and connects predictive analytics with the wider railway operations dashboard.

The integration process begins when railway service information and the required input features are provided to the backend demand prediction service. The service loads the saved Random Forest model and its categorical encoders to prepare the input data and generate passenger demand predictions. The predictions are then used to support demand analysis and seat allocation.

The seat allocation component uses predicted demand together with train capacity and passenger class information. The PuLP optimization model applies the configured allocation constraints and generates class wise allocation results. These results can be

presented through the dashboard alongside predicted demand and seat utilization information.

The integration also separates live prediction from offline experimentation. The Random Forest model is used by the live demand prediction service while the LSTM model is used for offline experimental analysis. This distinction ensures that the reported LSTM results are not incorrectly attributed to the live system.

The integration supports the following data flow:

Railway data → Backend demand prediction service → Random Forest model → Predicted passenger demand → Seat allocation optimization → Dashboard results

This design connects the forecasting and allocation stages while allowing each component to be evaluated independently. The integration evidence should be supported by screenshots of the live demand forecasting dashboard and the seat allocation interface.

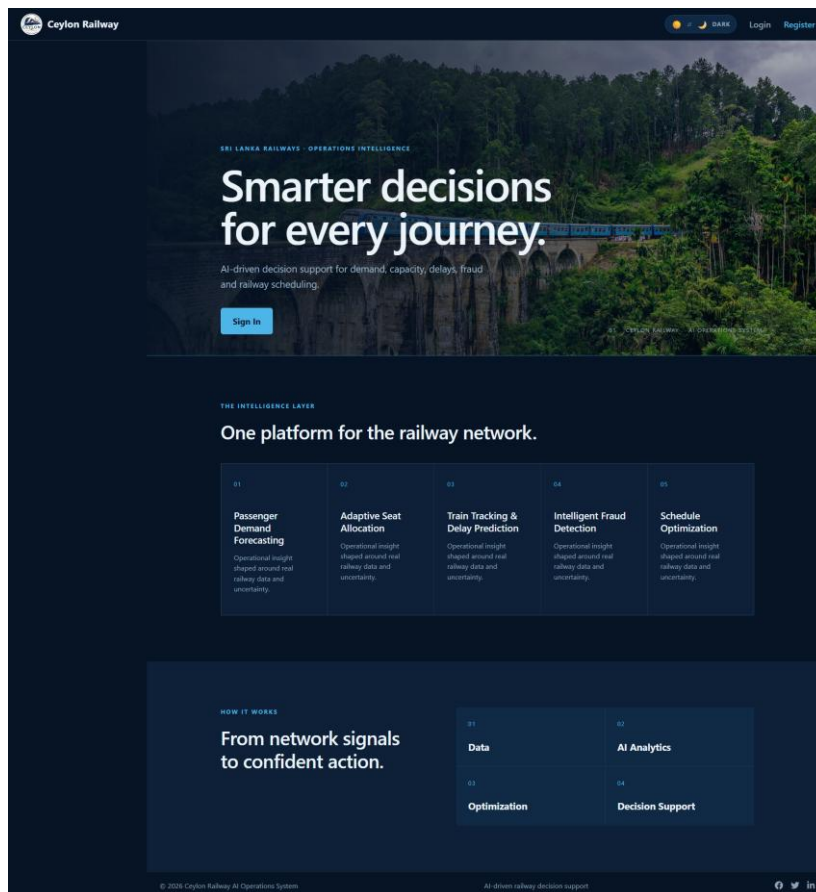


Figure 4 CeylonRail Intelligence System Home Page

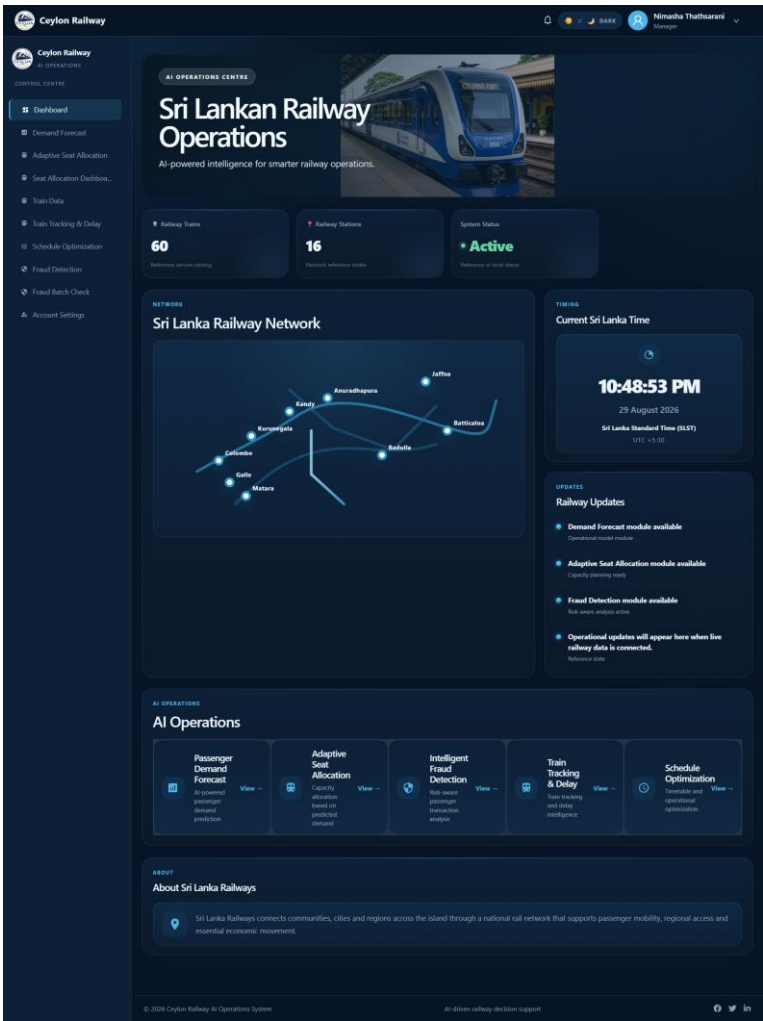


Figure 5:CeylonRail Intelligence Operations Dashboard

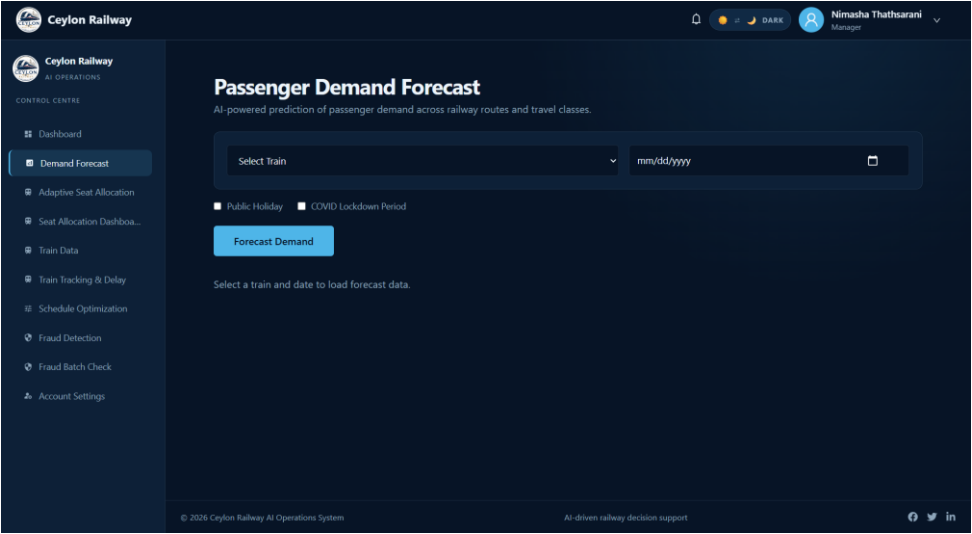


Figure 6:Passenger Demand Forecasting Interface

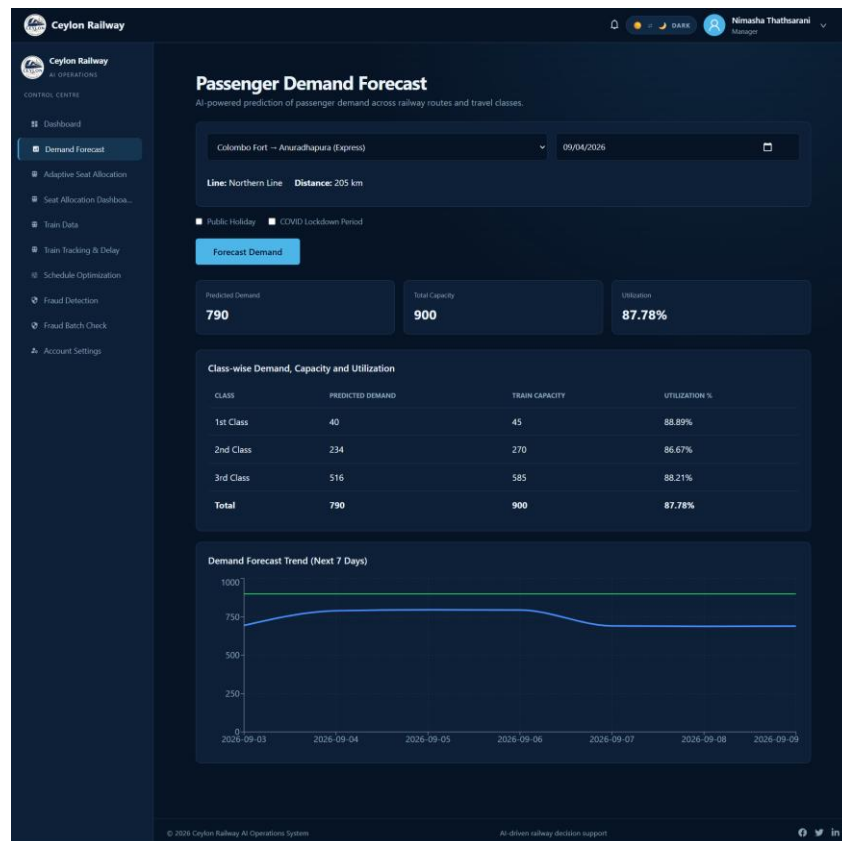


Figure 7: Passenger Demand Forecasting Results and Seven Day Demand Trend

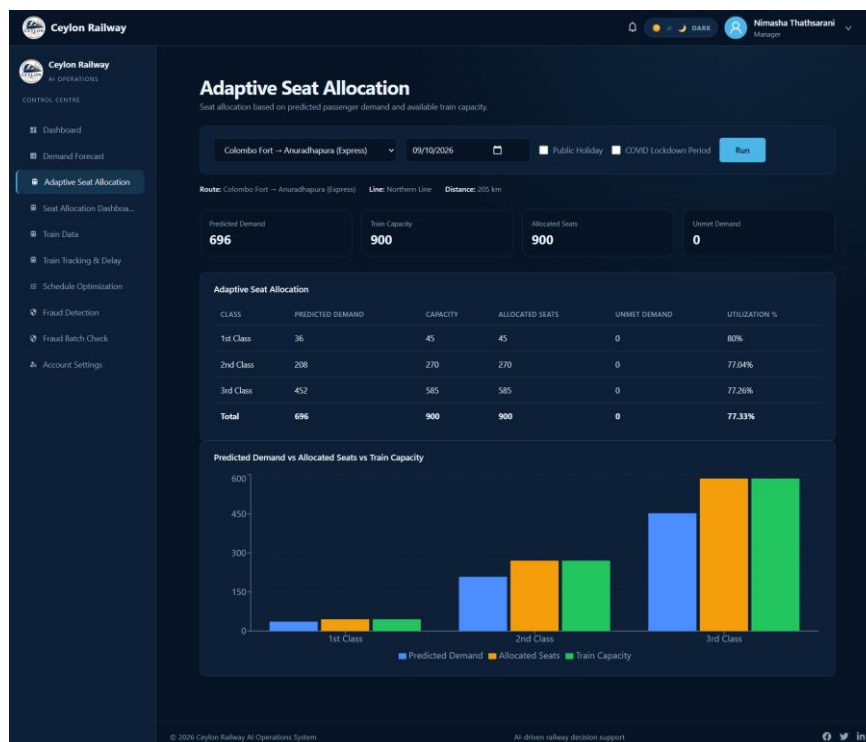


Figure 8: Adaptive Seat Allocation Interface and Allocation Results

5.4 Verification and software testing

The demand forecasting and adaptive seat allocation component was verified through model evaluation and inspection of the generated allocation results. Testing focused on forecasting accuracy, consistency of model inputs and compliance with the allocation constraints.

The forecasting models were evaluated using the 2024 test dataset. Random Forest achieved an MAE of 24.10, an RMSE of 37.67 and a MAPE of 13.12 percent. XGBoost achieved an MAE of 24.20, an RMSE of 38.13 and a MAPE of 13.43 percent. These results support the selection of Random Forest as the production forecasting model.

The seat allocation component was examined to determine whether the generated allocations satisfied the configured constraints. These include allocating the total available train capacity and keeping each passenger class within its specified minimum and maximum allocation limits. The generated results were also inspected to compare predicted demand with allocated seats and unmet demand.

The forecasting interface and adaptive seat allocation dashboard provide visual evidence of the implemented system. The forecasting interface accepts train and date inputs while the allocation interface presents predicted demand, train capacity, allocated seats and utilization values.

The testing evidence available for this thesis includes model evaluation results, class wise allocation outputs and screenshots of the implemented interfaces. However, a complete automated software testing report or formal test case execution record has not been established by the evidence reviewed so far. Therefore, the component should not be described as having passed comprehensive automated testing unless those test results are available.

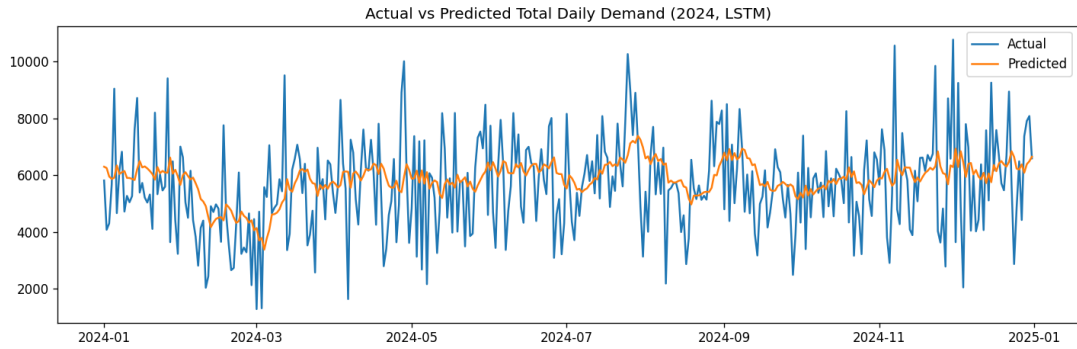


Figure 9: Demand forecasting vs adaptive seat allocation

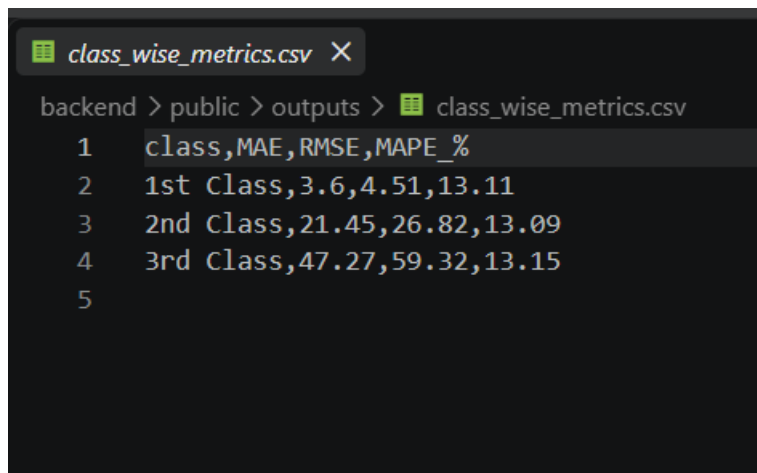


Figure 10: Forecasting metrics

5.5 Deployment and operation

The demand forecasting and adaptive seat allocation component is integrated into the CeylonRail Intelligence web application. The application provides a dashboard through which users can access demand forecasting and seat allocation functions.

The demand forecasting interface allows the user to select a railway service and travel date and specify relevant conditions such as public holidays and COVID lockdown periods. When forecasting is requested the system displays predicted passenger demand, train capacity and utilization information. The interface also presents class wise demand and capacity information and a seven day demand trend.

The adaptive seat allocation interface allows the user to select a route and date before running the allocation process. The results include predicted demand, total train

capacity, allocated seats and unmet demand. A class wise table and chart provide a comparison of predicted demand, allocated seats and available capacity.

The production forecasting service uses the saved Random Forest model and its associated categorical encoders. The separate LSTM model is used for offline experimental analysis and is not part of the live prediction service.

The current implementation demonstrates the operation of the forecasting and allocation interfaces within the web application. However, the available evidence does not establish that the system has been deployed in a live Sri Lanka Railways operational environment. It should therefore be described as an implemented research prototype rather than a production railway system.

6 RESULTS AND VALIDATION

6.1 Research results

The experiments evaluated passenger demand forecasting using Random Forest and XGBoost with a separate LSTM experiment to investigate sequential demand prediction. The research also examined historical demand patterns and implemented an integer optimization model for adaptive seat allocation.

The primary forecasting evaluation used historical railway data covering 2020 to 2024. Data from 2020 to 2023 was used for training and 2024 was retained for testing. Model performance was measured using Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE).

Table 6.1: Comparison of Primary Demand Forecasting Models

Table 11: Comparison of Primary Demand Forecasting Models

Model	MAE	RMSE	MAPE
Random Forest	24.10	37.67	13.12%
XGBoost	24.20	38.13	13.43%

Random Forest achieved the lowest error values across all three metrics. Its MAE was 24.10 compared with 24.20 for XGBoost while its RMSE and MAPE were also lower. Based on these experimental results, Random Forest was selected as the production forecasting model. Although the difference between the models was small, Random Forest provided the best overall result in the reported comparison.

A separate LSTM experiment was conducted using a 14 day lookback period. The model predictions were compared with actual daily demand during 2024. The resulting graph showed that the predicted series followed some broader demand patterns but did not capture every short term fluctuation in actual demand.

Table 12: Class Wise LSTM Forecasting Results

Passenger class	MAE	RMSE	MAPE
First class	3.60	4.51	13.11%
Second class	21.45	26.82	13.09%

Third class	47.27	59.32	13.15%
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The results indicate that the absolute prediction errors differed between passenger classes. Third class recorded the highest MAE and RMSE while first class recorded the lowest values. The MAPE values were similar across the three classes. These results should be interpreted in relation to the different demand scales of the passenger classes. They represent the LSTM experiment and should not be attributed to the production Random Forest model.

Exploratory analysis also revealed variations in passenger demand across time and passenger classes. The 30 day rolling average showed substantial changes during the earlier dataset period followed by generally higher demand levels during parts of 2022 and 2023. Monthly analysis showed differences in average demand between months, with December recording the highest average demand per record in the displayed analysis. Friday, Saturday and Sunday showed higher average demand than the other days of the week in the weekday analysis.

The holiday comparison showed higher average seat utilization during public holidays than during non holiday periods in the displayed dataset. These findings describe the patterns observed in the available data and do not establish that holidays or particular days directly cause higher demand.

6.2 Product and technical results

The implemented CeylonRail Intelligence component provides passenger demand forecasting and adaptive seat allocation through a web based interface. The demand forecasting page displays predicted passenger demand, train capacity, utilization and class wise demand information. The adaptive seat allocation page uses predicted demand to generate an allocation across the three passenger classes.

The allocation interface displays the predicted demand, available train capacity, allocated seats and unmet demand. It also provides a class wise table and a chart comparing predicted demand with allocated seats and train capacity.

In the demonstrated example for the Colombo Fort to Anuradhapura Express route, the system displayed a total predicted demand of 696 against a train capacity of 900.

The allocation results assigned 45 seats to first class, 270 seats to second class and 585 seats to third class. The interface reported zero unmet demand and a total utilization value of 77.33 percent.

These values describe the specific example displayed in the application and should not be interpreted as average performance across all routes or dates. The result demonstrates that the interface can present demand estimates and allocation decisions for a selected service.

The allocation implementation uses PuLP to maximize satisfied predicted demand subject to the configured capacity constraints. The total allocated seats must equal the available capacity while each passenger class must remain between 70 percent and 120 percent of its base capacity. These constraints limit unrestricted redistribution between classes.

The technical implementation also separates the production forecasting model from the additional LSTM experiment. Random Forest is used by the live demand prediction service while the LSTM model supports offline evaluation. This separation allows the production functionality to remain distinct from the experimental investigation.

6.3 User or stakeholder validation

The user interface of the CeylonRail Intelligence system was examined to assess the presentation of passenger demand forecasts and seat allocation results. The implemented interfaces allow users to select railway services and travel dates, view predicted passenger demand and examine the proposed distribution of seats across passenger classes.

The demand forecasting interface presents prediction results and supporting visualizations while the adaptive seat allocation interface displays predicted demand, allocated seats, available capacity and utilization information. These functions demonstrate the implementation of the intended user facing features.

However, formal user acceptance testing involving railway staff or passengers was not conducted as part of the available evaluation. Therefore, the usability and practical effectiveness of the system have not been validated through direct stakeholder

feedback. The current assessment is limited to the implemented interfaces and their displayed outputs.

Future work should include structured usability testing with relevant railway stakeholders to evaluate the clarity of forecasting results, the usefulness of allocation recommendations and the suitability of the system for railway operational decision making.

6.4 Integration results

The demand forecasting and adaptive seat allocation components were integrated into the CeylonRail Intelligence system to support railway operational decision making. The forecasting component provides predicted passenger demand, while the allocation component uses demand predictions and train capacity constraints to calculate a proposed distribution of seats across passenger classes.

The integrated application provides interfaces for viewing demand forecasts, examining demand trends and reviewing seat allocation results. The allocation interface presents predicted demand, allocated seats, available capacity and seat utilization information, allowing users to examine the relationship between expected demand and available seating capacity.

The implemented system demonstrates the functional connection between demand forecasting and seat allocation. However, comprehensive integration testing involving all system components under real railway operating conditions was not established in the available evidence. Therefore, the results demonstrate implemented integration functionality rather than verified operational performance in a live railway environment.

6.5 Summary of findings

The results demonstrate the feasibility of applying machine learning based demand forecasting and optimization techniques to railway passenger demand management. The Random Forest model achieved an MAE of 24.10, an RMSE of 37.67 and a MAPE of 13.12% on the reported test dataset. It performed slightly better than the evaluated

XGBoost model and was selected as the primary model for the demand forecasting service.

A separate LSTM experiment was conducted to examine aggregate daily demand forecasting. Its class wise evaluation results showed different error magnitudes across First Class, Second Class and Third Class, indicating that forecasting performance varied by passenger class.

The seat allocation component used integer optimization to distribute available capacity according to predicted demand and predefined allocation constraints. The web application brought forecasting and allocation outputs together through user facing interfaces.

Nevertheless, the findings should be interpreted in light of the available dataset evidence and evaluation scope. Formal stakeholder validation and comprehensive real world operational testing were not established. The results therefore support the technical feasibility of the proposed approach but do not establish its effectiveness in live railway operations

7 DISCUSSION

7.1 Interpretation of findings

The findings indicate that machine learning can support railway passenger demand forecasting and that forecasting results can be used as inputs for adaptive seat allocation. The Random Forest model achieved an MAE of 24.10, an RMSE of 37.67 and a MAPE of 13.12% on the reported test dataset. These results indicate that the model produced useful demand estimates within the scope of the conducted evaluation. However, the errors also show that predicted passenger demand does not always match actual demand.

The Random Forest model performed slightly better than XGBoost across the reported evaluation metrics. Its lower MAE, RMSE and MAPE supported its selection as the primary model for the demand forecasting service. Nevertheless, these results do not establish that Random Forest will outperform other forecasting approaches across different routes, periods or demand conditions.

The separate LSTM experiment examined aggregate daily passenger demand and provided additional evidence of the potential of sequence based forecasting. Its class wise results showed that prediction errors differed between passenger classes. Third Class recorded a higher MAE and RMSE than First Class and Second Class. This difference should be interpreted in relation to the scale and variability of demand in each class rather than as evidence that the model is necessarily less suitable for Third Class.

The adaptive seat allocation component used predicted demand, train capacity and predefined class allocation constraints to determine a proposed seat distribution. This demonstrates how demand forecasting can inform capacity allocation. However, the implementation alone does not establish that adaptive allocation improves passenger satisfaction or seat utilization compared with a static allocation policy. Such conclusions require a controlled comparison using consistent demand scenarios and clearly defined evaluation measures.

Overall, the findings support the technical feasibility of connecting machine learning based demand forecasting with optimization based seat allocation. Further evaluation

using verified railway data, broader testing across routes and seasons and comparisons with alternative allocation policies would be necessary before drawing conclusions about operational effectiveness.

7.2 Comparison with literature baselines and existing solutions

Previous research has demonstrated the value of machine learning and deep learning techniques for forecasting passenger demand in transportation systems. Statistical approaches such as ARIMA and more advanced methods such as LSTM have been explored for modelling temporal patterns. Research on transportation demand forecasting has also investigated the use of spatial and temporal information to improve predictions across routes and time periods.

The proposed CeylonRail Intelligence component follows this direction by using historical railway ticketing information, route characteristics, passenger class information and temporal features to estimate demand. The Random Forest model was selected after comparison with XGBoost using MAE, RMSE and MAPE. The results showed a small performance advantage for Random Forest on the evaluated dataset.

Unlike forecasting approaches that focus only on predicting passenger demand, this component also connects demand estimates to an optimization based seat allocation process. The allocation engine considers train capacity and class specific allocation limits when determining the proposed distribution of seats. This provides a practical connection between demand prediction and operational decision support.

The project also includes a separate LSTM experiment for aggregate daily demand forecasting. However, the available results do not establish a direct, controlled comparison between the LSTM and Random Forest models under identical prediction targets and evaluation conditions. Therefore, it would not be appropriate to conclude that one model is universally superior to the other.

Similarly, although previous research has examined fairness aware allocation and the efficient use of transport capacity, the current implementation does not provide sufficient comparative evidence to establish that its allocation strategy outperforms existing railway allocation methods. Its contribution is the integration of demand

forecasting and constrained seat allocation within a Sri Lankan railway focused application.

The comparison therefore indicates that the proposed component applies established forecasting and optimization techniques within a locally focused railway system. Its potential advantages are the combination of route and class specific demand estimation with capacity constrained allocation. Further controlled experiments are required to quantify improvements in forecast accuracy, seat utilization and passenger fairness relative to established alternatives.

7.3 Research contribution

The main research contribution of this project is the development of a Sri Lankan railway focused approach that connects passenger demand forecasting with adaptive seat allocation. Rather than treating forecasting and allocation as separate tasks, the proposed component uses predicted demand as an input to an optimization process that considers train capacity and passenger class constraints.

The component contributes in three main ways. First, it applies machine learning techniques to historical railway ticketing data to estimate passenger demand using temporal, route and service characteristics. Second, it investigates different forecasting approaches through the evaluation of Random Forest and XGBoost, with an additional LSTM experiment for aggregate daily demand. Third, it implements an integer optimization based allocation engine to determine seat distributions according to predicted demand and predefined operational constraints.

The work also provides a technical foundation for examining the relationship between forecast uncertainty, capacity utilization and the distribution of seats among passenger classes. However, the current evaluation does not quantitatively establish improvements in fairness or utilization compared with a baseline allocation policy.

Therefore, the research contribution is primarily the implementation and integration of a demand forecasting and constrained seat allocation approach within the CeylonRail Intelligence system. Its effectiveness under real railway operating conditions remains an area for further investigation.

7.4 Product contribution

The product contribution of this research is the implementation of the demand forecasting and adaptive seat allocation component within the CeylonRail Intelligence web application. The component provides railway demand insights and allocation recommendations through a user interface designed to support operational decision making.

The demand forecasting interface allows users to select railway services and travel dates and examine predicted passenger demand alongside supporting visualizations. The system also presents demand trends and analytical summaries that help users explore variations in passenger demand over time.

The adaptive seat allocation interface presents predicted demand, allocated seats and seat utilization information for passenger classes. Its optimization engine uses predefined capacity and allocation constraints to generate a proposed distribution of seats. This connects the forecasting output to a practical allocation function within the application.

The implementation demonstrates how machine learning models and mathematical optimization can be combined in a single railway focused decision support application. It provides a foundation for future improvements such as incorporating live ticketing information, evaluating additional allocation policies and integrating verified operational data.

However, the current product should be regarded as a research prototype rather than a production ready railway management system. Further security assessment, operational testing, stakeholder evaluation and validation with real railway data would be necessary before deployment in an operational environment.

7.5 Limitations and threats to validity

- **Dataset limitations:** The forecasting models were evaluated using the available railway ticketing dataset covering January 2020 to December 2024. The provenance and representativeness of the dataset have not been fully established. Therefore, the results may not reflect actual passenger demand patterns across all Sri Lankan railway services. Changes in travel behaviour,

service schedules and seasonal conditions may also affect forecasting performance.

- **Forecasting limitations:** The Random Forest and XGBoost models were evaluated using a chronological training and testing split. Although this approach supports evaluation on a later period, the reported metrics alone do not establish consistent performance across different routes, passenger classes and demand conditions. The separate LSTM experiment also used a different prediction target, limiting direct comparison with the primary forecasting model.
- **Feature engineering limitations:** The forecasting approach uses historical demand and temporal information to identify patterns. However, the reliability of lag and rolling features depends on how historical observations are ordered and processed. Any preprocessing that uses information from future observations could introduce data leakage and affect the validity of evaluation results. Further verification of feature generation is therefore necessary.
- **Seat allocation limitations:** The optimization engine applies predefined capacity and class allocation constraints to generate seat distributions. The current evaluation does not establish whether these constraints reflect all practical railway operating requirements. Furthermore, a controlled comparison against a static allocation baseline has not been established, so improvements in seat utilization, demand satisfaction and fairness cannot be confirmed.
- **Validation limitations:** Formal user acceptance testing with railway staff or passengers was not conducted as part of the available evaluation. The application interfaces demonstrate implemented functionality but do not establish usability or operational suitability. Testing under live railway conditions was also not established.

7.6 Implications

- **Operational implications:** Passenger demand forecasts can help railway operators anticipate variations in ticket demand across travel dates, routes and passenger classes. When combined with seat allocation optimization, these

forecasts can support decisions about distributing available seating capacity according to expected demand.

- **Technical implications:** The integration of machine learning and integer optimization demonstrates a practical approach to connecting demand prediction with resource allocation. The modular design also provides a foundation for incorporating additional forecasting models, operational data and allocation constraints in future versions.
- **Passenger implications:** A demand informed allocation approach has the potential to improve the use of available seating capacity and support more balanced seat distribution across passenger classes. However, these benefits require further evaluation through controlled experiments and stakeholder feedback before they can be confirmed.
- **Research implications:** The project provides a starting point for further investigation into railway demand forecasting using Sri Lankan railway data. Future studies could examine model performance across individual routes, incorporate verified weather and event information and compare alternative allocation strategies using consistent evaluation metrics.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research focused on developing an AI driven railway passenger demand forecasting and adaptive seat allocation component for the CeylonRail Intelligence system. The main objective was to investigate how machine learning techniques and optimization methods could be combined to support passenger demand analysis and the distribution of available seating capacity within the Sri Lankan railway context.

The implemented forecasting component used historical railway ticketing data and relevant temporal and operational features to estimate passenger demand. Random Forest and XGBoost models were evaluated using MAE, RMSE and MAPE. Random Forest achieved an MAE of 24.10, an RMSE of 37.67 and a MAPE of 13.12% on the reported test dataset and was selected as the primary forecasting model. A separate LSTM experiment was also conducted to investigate aggregate daily demand forecasting.

The adaptive seat allocation component used integer optimization to generate proposed seat distributions based on predicted demand, train capacity and predefined class allocation constraints. These functions were integrated into the web application, allowing users to examine demand forecasts and allocation results through dedicated interfaces.

The findings demonstrate the technical feasibility of connecting machine learning based demand forecasting with optimization based seat allocation. However, the evaluation does not establish improved performance against a static allocation baseline or validate the system under real railway operating conditions. Further testing with verified operational data and feedback from railway stakeholders is required before the system can be considered suitable for practical deployment.

8.2 Achievement of objectives

Objective	Achievement
Collect, integrate and preprocess railway ticketing data and relevant features	A railway ticketing dataset covering 2020 to 2024 was processed and used to

	construct features for demand forecasting. The provenance and representativeness of the dataset require further verification.
Develop and evaluate demand forecasting models	Random Forest and XGBoost were evaluated using MAE, RMSE and MAPE. A separate LSTM experiment was conducted for aggregate daily demand forecasting. The full set of models originally proposed in the TAF was not implemented in the available evidence.
Formulate and implement adaptive seat allocation	An integer optimization based allocation engine was implemented using predicted demand, train capacity and predefined class allocation constraints.
Compare adaptive allocation against static allocation under varying demand scenarios	The allocation functionality was implemented, but a comprehensive controlled comparison against static allocation was not established in the available results. The proposed comparison therefore remains incomplete.

8.3 Individual research and product contributions

The individual research contribution is the implementation and evaluation of a railway passenger demand forecasting approach using machine learning, together with an optimization based method for allocating seats across passenger classes. The work examines how historical demand patterns and operational features can inform demand

predictions and how those predictions can be incorporated into capacity constrained allocation.

The individual product contribution is the development of the forecasting and adaptive seat allocation functionality within the CeylonRail Intelligence web application. This includes the primary Random Forest forecasting service, the separate LSTM experiment, the integer optimization based allocation engine and interfaces for displaying demand forecasts and allocation results.

The integration of these functions provides a research prototype for exploring the relationship between passenger demand prediction and railway seating capacity management. The current contribution is primarily technical and experimental, as formal stakeholder validation and comprehensive real world operational evaluation have not been established.

8.4 Future directions

1. Improve data quality and coverage: Validate the provenance of the existing dataset and incorporate reliable operational ticketing records covering additional routes, passenger classes and travel periods.
2. Strengthen forecasting evaluation: Implement and compare additional proposed models such as ARIMA, Prophet, GRU and TCN where appropriate. Evaluate models using consistent data splits, prediction targets and metrics.
3. Improve feature engineering: Verify that lag and rolling features use only information available at prediction time. Apply leakage prevention measures and test the models across different routes and demand conditions.
4. Evaluate allocation performance: Conduct controlled experiments comparing adaptive allocation with static allocation under consistent demand scenarios. Measure seat utilization, demand satisfaction, booking rejection rates and clearly defined fairness metrics.
5. Enhance allocation constraints: Investigate realistic railway operating requirements and improve the allocation model to account for practical restrictions and passenger needs.

6. Conduct stakeholder validation: Obtain feedback from railway staff and potential users through structured usability testing and assess the clarity and usefulness of forecasts and allocation recommendations.
7. Investigate operational deployment: Explore integration with verified railway information systems, appropriate security controls and reliable data update mechanisms before considering deployment in a live environment.

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GLOSSARY

Term	Definition
AI (Artificial Intelligence)	Technologies that enable computer systems to perform tasks requiring human like intelligence.
Adaptive Seat Allocation	A method of distributing available railway seats based on predicted passenger demand and predefined allocation constraints.
LSTM (Long Short Term Memory)	A recurrent neural network architecture designed to learn temporal dependencies in sequential data.
MAE (Mean Absolute Error)	An evaluation metric that measures the average absolute difference between actual and predicted values.
MAPE (Mean Absolute Percentage Error)	An evaluation metric that expresses the average absolute prediction error as a percentage of actual values.
Machine Learning	A branch of artificial intelligence that enables systems to learn patterns from data and generate predictions.
Random Forest	An ensemble machine learning algorithm that combines multiple decision trees to produce predictions.
RMSE (Root Mean Squared Error)	An evaluation metric that measures the square root of the average squared difference between actual and predicted values.
Seat Utilization	The proportion of available seating capacity used or occupied during a specified period.
Spatio Temporal Forecasting	A forecasting approach that considers spatial relationships and changes over time.
XGBoost	An optimized gradient boosting framework that builds an ensemble of decision trees for predictive tasks.
Integer Optimization	An optimization method in which some or all decision variables are restricted to integer values.
Passenger Demand Forecasting	The process of estimating future passenger demand using historical information and relevant explanatory features.

APPENDICES

Appendix A Research instruments or extended results

This appendix presents supplementary research materials and extended results that support the individual research component, *Advanced Spatio Temporal Demand Forecasting and Adaptive Seat Allocation using AI*.

Include the following materials where relevant:

- Extended Random Forest and XGBoost evaluation results.
- Class wise LSTM forecasting metrics.
- Additional demand forecasting charts and exploratory data analysis.
- Supplementary seat allocation results and optimization outputs.
- Additional tables or technical evidence that would interrupt the main discussion if included in the chapters.

Appendix B Academic integrity and AI use disclosure

This appendix contains the academic integrity declaration and AI use disclosure required by the university.

Include the university's required declaration form or template and accurately disclose any AI tools used during the research and thesis preparation, in accordance with university policy. Clearly distinguish your own research, implementation and analysis from any AI assisted writing or editing.